A fixed-target approach to the sharp iid Berry–Esseen inequality*

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Abstract

We study the conjectured sharp iid Berry–Esseen constant by maximizing excess over its fixed target. Attainment and contact give a common moment-defect budget; a specified Fourier certificate supplies coarse entry. For $n \geq 700$, direct geometry and two contact rays acquire one support strip. A single-count fourth-moment bound handles small noise, and a five-count identity handles the rest. Both comparisons preserve the actual marginal, threshold atoms and complete tails. The resulting global implication remains conditional on the stated entry predicates and a genuine finite base through 699, which is not proved here.

1 The sharp constant and the proof route

For centered, variance-one laws P with $\rho(P) = \mathbb{E}|X|^3 < \infty$, let S_n be a sum of independent copies and set

$$\Delta_n(P) = \sup_{H\in\mathbb{R}} |\mathbb{P}_P(S_n \leq H) - \Phi(H/\sqrt{n})|, \qquad C_n = \sup_P \frac{\sqrt{n}\Delta_n(P)}{\rho(P)}.$$

The iid sharp-constant conjecture is

$$\sup_{n\geq 1} C_n = C_{\mathrm{E}}, \qquad C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}. \tag{1}$$

Here Φ, ϕ denote the standard normal distribution function and density; H is always a raw sum threshold. Finite decimals mean exact rationals. Affine standardization gives the usual arbitrary-mean, positive-variance form.

The classical Berry–Esseen theorem gives the order $n^{-1/2}$ [1, 2]. Esseen's fixed Bernoulli example gives the sharp lower approximation [3]. Namely, for $p_* = (4-\sqrt{10})/2$, $q_* = 1-p_*$ and $X = (B-p_*)/\sqrt{p_*q_*}$,

$$\lim_{n\to\infty} \frac{\sqrt{n}\Delta_n(P)}{\rho(P)} = \frac{3+q_*-p_*}{6\sqrt{2\pi}(p_*^2+q_*^2)} = C_{\mathrm{E}}. \tag{2}$$

The last equality uses $p_*^2 + q_*^2 = 10 - 3\sqrt{10}$. Thus one fixed two-point law already makes every constant smaller than C_{E} fail for sufficiently large n. The remaining task is the upper bound, uniformly over both the marginal and the index. Strict inequalities at each finite

*This paper was written with substantial assistance from AI systems; see the end of Section 8.

index are compatible with this limiting lower bound. The universal upper conjecture is associated with Zolotarev [6]. We use Shevtsova's structural moment bound [5] and Schulz's strict finite-binomial theorem [4], both stated where applied.

Fix $c = C_{\mathrm{E}}$ and maximize the excess

$$a(P, H) = \sqrt{n}\{\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho(P). \tag{3}$$

The choice of objective matters. Its target c is fixed before the law is varied, so the same coefficient appears in the variational inequality, the conservation identity and the predecessor estimate. The third-moment penalty also controls escape of mass in a maximizing sequence. These two features let us study an attained law and then extract information from its support.

The fixed moment penalty gives attainment and compact-support contact. Integrating contact against the zero-bias density gives one budget for excess, moment defect and leakage. Complete coarse Fourier entry supplies $nD \le 32.20665378$. At $n \ge 700$, direct geometry and two contact rays then acquire one probability/support strip. Two comparisons, split at noise $\lambda = 1/100$, exclude its nonnegative excess.

All steps retain the same actual marginal. Reflection chooses its positive closed-event orientation before signed acquisition. The local comparison itself requires neither contact nor a predecessor; these are used only to acquire its hypotheses. The global implication in Section 7 still needs a genuine base through 699: a base only through 41 leaves $42, \dots, 699$ unresolved.

1.1 Reading the proof

The argument follows one marginal through four stages. The distinction between acquiring hypotheses and applying the resulting comparison is useful when reading the longer calculations.

1. Section 2 selects an attained fixed-target maximum and derives contact and conservation. The support becomes compact before fourth and higher moments enter.

2. Sections 3 and 4 turn conservation into coarse bounds for its physical gap defect and raw threshold. The entry includes explicit finite certificate conditions; Appendix A supplies their analytical Fourier bounds.

3. Section 5 uses those coarse bounds to acquire one support strip. Its moment geometry is independent of contact; the two outer support rays use contact and the actual sharp predecessor.

4. Section 6 compares every law in that strip with a binomial law. It needs only the strip hypotheses. Section 7 then states precisely how this local exclusion composes with a finite base.

There are two coordinate systems. The physical random variable X is centered and has variance one. Its sign indicator is $B_0 = \mathbf{1}_{\{X>0\}}$, with $p = \mathbb{P}(B_0 = 1)$ and $q = 1 - p$. Subtract the conditional means and divide by their separation ℓ to obtain the count noise U. The exact relation is

$$X = \ell(B_0 + U - p), \qquad \mathbb{E}(U \mid B_0) = 0.$$

The following quantities will recur; each is defined again where used.

θ	$\mathbb{E}\{X - \mathbb{E}(X \mid B_0)\}^2$
s	$\mathbb{E}U^2 = \theta/\ell^2$
λ	ns
a_0, a_1	inward support displacements in count coordinates
D	$(3+\sqrt{10})\rho - \gamma - 3J$

Here $\gamma = \mathbb{E}X^3$, $\beta = \mathbb{E}[X|X|]$, and J is the actual weighted support-gap sum. The letters $d = a_0 + a_1$ and D have different roles: d measures inward displacement, whereas D is the physical moment defect. The normalization satisfies $pq + s = \ell^{-2}$, so reconstruction leaves both the actual law and the raw threshold H unchanged.

Reflection chooses the positive closed-event orientation before entry. All subsequent signed quantities are acquired in that orientation. Alternative smaller-index routes and wider comparison domains are preserved in the repository; they are not extra premises of this selected route.

2 Fixed-target extremality and conservation

We maximize excess above a fixed target c, retaining that excess through contact and conservation. Sums have independent identically distributed summands with mean zero, variance one and finite third absolute moment ρ. Fix $n \geq 2$, $c > 0$, and a real raw sum threshold H. Set

$$z = H/\sqrt{n}, \quad F = \mathbb{P}(S_n \leq H), \quad a = \sqrt{n}\{F - \Phi(z)\} - c\rho,$$
$$\varphi = \phi(z), \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3.$$

In particular a is not divided by ρ.

2.1 Attainment, contamination and the entire closed support

Proposition 2.1. *For each fixed n and $c > 0$, the supremum of a over these laws and thresholds is attained, and the closed-CDF discrepancy at every attaining pair is positive. At every maximizing pair,*

$$I(y) = n^{3/2}\{h(y) - F\} + ny\varphi + \tfrac{1}{2}H\varphi(y^2 - 1) + \tfrac{3}{2}c\rho(y^2 - 1) - c\{|y|^3 - \rho - 3y\beta\} \leq 0, \qquad h(y) = \mathbb{P}(S_{n-1} \leq H - y), \tag{4}$$

with equality on the whole closed support. That support is compact. Neither statement assumes $a \geq 0$ or $H \geq 0$.

Proof. Write A for the supremum. Since $\rho \geq 1$ and the discrepancy is at most one, $A \leq \sqrt{n} - c$. Symmetric Bernoulli evaluated at $H = n$ gives $A \geq \sqrt{n}(1 - \Phi(\sqrt{n})) - c > -c$. For a maximizing sequence, $a_j \to A$ and $c\rho_j \leq \sqrt{n} - a_j$ give a common third-moment ceiling. The discrepancies satisfy $\sqrt{n}(F_j - \Phi(H_j/\sqrt{n})) = a_j + c\rho_j \geq a_j + c > \eta$ eventually for some $\eta > 0$. Chebyshev bounds the negative threshold tail by n/H_j^2; the positive discrepancy is at most the normal tail, also at most n/H_j^2. Thus H_j is bounded.

Tightness gives a subsequence $P_j \Rightarrow P$, $H_j \to H$ and $\rho_j \to b$. The common third-moment ceiling M gives tail bounds $\int_{|x|>K} x^2 dP_j \leq M/K$ and $\int_{|x|>K} |x| dP_j \leq M/K^2$. Bounded

continuous truncations therefore preserve mean and variance. Lower semicontinuity gives $1 \le \rho(P) \le b$. For fixed n, the sum laws converge weakly. Applying the closed-set inequality first at $H + t$ and then letting $t \downarrow 0$ gives $\limsup F_j(H_j) \le F_P(H)$, including an atom at H. Consequently

$$a(P, H) \ge A + c\{b - \rho(P)\} \ge A.$$

Maximality forces equality and $\rho(P) = b$. Convergence of third moments is a conclusion here; it was not inferred from bounded third moments. The discrepancy remains positive since $A + c\rho \ge A + c > 0$.

For contact, contaminate the raw marginal by $(1 - \epsilon)P + \epsilon\delta_y$. Its mean is ϵy, variance $v_\epsilon = 1 + \epsilon(y^2 - 1) - \epsilon^2 y^2$, and centered third absolute moment has derivative $|y|^3 - \rho - 3y\beta$ at zero. The last assertion follows by dominated differentiation of $|x - \epsilon y|^3$, with bound $3|y|(|x| + |y|)^2$. Evaluate the standardized law at the standardized image of the fixed raw event $S_n \le H$. The raw event probability is a polynomial in ϵ, with derivative $n(h(y) - F)$ even at an atom collision. The normal argument is $(H - n\epsilon y)/\sqrt{nv_\epsilon}$, whose derivative is $-\sqrt{n}y - H(y^2 - 1)/(2\sqrt{n})$. The standardized third absolute moment is the centered third moment multiplied by $v_\epsilon^{-3/2}$. Its derivative is therefore

$$|y|^3 - \rho - 3y\beta - \tfrac{3}{2}\rho(y^2 - 1).$$

Combining this with the two event derivatives gives precisely $I(y)$. Every perturbed, standardized marginal is admissible for sufficiently small $\epsilon \ge 0$, so maximality makes this right derivative nonpositive.

Integrating against P gives zero: $\mathbb{E}h(X) = F$, $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$ and $\mathbb{E}|X|^3 = \rho$ cancel the terms. They are all integrable. The closed-event map h is upper semicontinuous; at an atom it takes the larger one-sided value. Thus I is upper semicontinuous. Strict negativity at a closed-support point would hold with a uniform negative margin on a neighborhood of positive P-mass, contradicting its zero integral. Finally $I(y) \le -c|y|^3 + C_2 y^2 + C_1|y| + C_0$ for constants depending on this pair. It is strictly negative on both sufficiently distant rays, proving compact support. No higher moments were used before this step. □

2.2 Reflection, equality, and the first unresolved index

The sign of the attained maximum detects failure of a strict row bound, including equality. If C_n denotes the supremum of the normalized absolute discrepancy, reflection gives, including an atom at H,

$$\mathbb{P}(-S_n \le -H) - \Phi(-z) = \Phi(z) - \mathbb{P}(S_n < H) \ge \Phi(z) - F.$$

Thus each negative closed-CDF error is bounded by a positive closed-CDF error of the reflected law; equality at an atom is not assumed. Consequently C_n is also the supremum of positive closed scores r. Since $a = \rho(r - c)$, $C_n < c$ gives $A \le C_n - c < 0$, and $C_n > c$ gives $A > 0$. If $C_n = c$, choose $r_j \uparrow c$ with $r_j \ge c/2$. Then $\rho_j \le 2\sqrt{n}/c$, so $\rho_j(r_j - c) \to 0$ and $A = 0$. Hence $C_n \ge c$ supplies an attained pair with $a \ge 0$.

Corollary 2.2 (First failure of a strict bound)**.** *Assume a genuine finite base $C_j \le c$ for $1 \le j < N$. If a strict bound $C_n < c$ fails at any index $n \ge N$, there is a least such index, an attained fixed-target pair with $a \ge 0$, and true predecessor bounds $C_j \le c$ for every $j < n$. The bounds are strict for $N \le j < n$.*

Proof. The nonempty set of failing indices at least N has a least member. The preceding equivalence supplies its maximizing pair. Indices below N use the finite base; the other earlier indices use minimality. □

Entry will use the true predecessor $C_{n-1} \le c$, without an upper bound on the maximizing score $c + a/\rho$.

Equivalently, $\Delta = Q - h \ge 0$ everywhere and $\Delta = 0$ on the closed support, where

$$\begin{aligned} Q(y) = F + \frac{H\varphi + c\rho}{2n^{3/2}} - \frac{n\varphi + 3c\beta}{n^{3/2}} y \\ - \frac{H\varphi + 3c\rho}{2n^{3/2}} y^2 + \frac{c}{n^{3/2}} |y|^3. \end{aligned} \tag{5}$$

2.3 Zero-bias integration and the exact identity

We now allow an arbitrary compactly supported standardized law, retaining the definitions of a, Q, h, Δ above without assuming contact. Put $\nu = \mathcal{L}(H - S_{n-1})$ and $\tau(t) = \mathbb{E}[X \mathbf{1}_{X \ge t}]$. Then $\tau \ge 0$, $\int \tau = 1$. Over the bounded complementary intervals (u, v) of the closed support K, define

$$J = \sum \tau_{u,v} (v-u)^2, \qquad J_1 = \sum \tau_{u,v} \int_u^v (t-u) t \, dt,$$
$$J_2 = \sum \tau_{u,v} \int_u^v (t-u) t |t| \, dt, \qquad L = \sum \tau_{u,v} \int_{[u,v)} (t-u) \, d\nu(t).$$

Here $\tau_{u,v}$ is the constant interior value. Introduce two contact residuals:

$$\xi = \mathbb{E}[X \Delta(X)], \qquad B = \int_K \tau(t) \Delta(t) \, dt + \sum_{(u,v)} \tau_{u,v} (v-u) \Delta(v). \tag{6}$$

The integral over K is a Lebesgue integral, not an expectation under P. It must be retained when the closed support has positive Lebesgue measure. If $K \subset [-R, R]$, then $J \le 2R$, $|J_1| \le RJ/2$, $|J_2| \le R^2 J/2$, and $0 \le L \le 2R$. Also Δ is bounded on the convex hull, so the sum in B is absolutely convergent because $\sum \tau_{u,v}(v-u) \le 1$.

Let $M_4 = \mathbb{E}X^4$, $M_5 = \mathbb{E}|X|^5$, and put

$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \qquad K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2,$$

$$\Omega = J_1 + (M_4 - 3)/6, \quad \varepsilon = \mathbb{E} q_z(S_n/\sqrt{n}) - \mathbb{E} q_z(Z_0) - 2\gamma\varphi/(3\sqrt{n}), \quad q_z(x) = (x-z)|x-z|,$$

where Z_0 is standard normal. Define $E_n = -n^{3/2}\varepsilon/2$ and

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \varphi\Omega) - H^2(c\rho + \gamma\varphi/2) + E_n. \tag{7}$$

Theorem 2.3 (Conservation with contact residuals)**.** *For every such law, every $H \in \mathbb{R}$ and every $c > 0$,*

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2} L = \mathcal{R}_c + n^{3/2}(B - H\xi). \tag{8}$$

At a fixed-target maximizing pair, $B = \xi = 0$. In particular,

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2} L = \mathcal{R}_c. \tag{9}$$

The coefficient of the unnormalized excess is $n + H^2$.

Proof. There are two direct evaluations of $\mathcal{A} = \int \tau\Delta$. Since $h(t) = \nu([t,\infty))$, one has $h(t) = h(v) + \nu([t,v))$ inside a gap. Thus bounded Fubini gives

$$\int_u^v \Delta(t)dt = (v-u)\Delta(v) + \int_u^v (t-u)(-Q'(t))dt - \int_{[u,v)} (t-u)d\nu(t).$$

An atom at v belongs to $h(v)$ and is not in the last integral; an atom at u is included but multiplied by zero. Summing, including the support integral in (6), gives

$$\mathcal{A} = \frac{\varphi J}{2\sqrt{n}} + \frac{3c\beta J/2 + (H\varphi + 3c\rho)J_1 - 3cJ_2}{n^{3/2}} - L + B. \tag{10}$$

For the second evaluation, write $g(X) - g(0) = \int_0^X g'(t)dt$. Multiply by X, take expectations separately on each sign, and use centering on the negative half-line. Fubini then gives $\int \tau g' = \mathbb{E}[Xg(X)]$ for absolutely continuous g with bounded derivative on the relevant compact range. In particular

$$\int t\tau = \gamma/2, \quad \int t^2\tau = M_4/3, \quad \int |t|^3\tau = M_5/4.$$

Apply that same identity conditionally to $g_s(x) = \min(s+x, H)$ and then use exchangeability. The choice of derivative at its single corner is immaterial under the density τ. It follows directly that

$$\int \tau(t)h(t)dt = \mathbb{E}[X_n \min(S_n, H)] = \frac{1}{n}\mathbb{E}[S_n \min(S_n, H)] = T,$$

where $T = \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{Z\le z}]$, $Z = S_n/\sqrt{n}$. The last expression vanishes at an endpoint atom. Integrating (5) against τ therefore yields

$$\mathcal{A} = F - \frac{\gamma\varphi}{2\sqrt{n}} + \frac{H\varphi(3-M_4)/6 + cK_q}{n^{3/2}} - T. \tag{11}$$

Set $M = \mathbb{E}[Z\mathbf{1}_{Z\le z}]$. Since $\mathbb{E}[Xh(X)] = M/\sqrt{n}$, integration of $XQ(X)$ against P gives

$$M + \varphi = -\frac{\gamma H\varphi}{2n} + \frac{cK_1}{n} - \sqrt{n}\xi. \tag{12}$$

The pointwise identity $q_z(x) = (x-z)^2 - 2(x-z)^2\mathbf{1}_{x\le z}$, with the same identity for a normal variable, gives

$$T - \Phi(z) = -\varepsilon/2 - \gamma\varphi/(3\sqrt{n}) + z(M+\varphi) - z^2(F - \Phi(z)).$$

Inserting (12) and $\sqrt{n}(F - \Phi(z)) = c\rho + a$ into (11) gives

$$\begin{aligned}\sqrt{n}\mathcal{A} &= c\rho + a - \gamma\varphi/6 + P_0/n + \sqrt{n}\varepsilon/2 + H\sqrt{n}\xi,\\ P_0 &= H^2(c\rho + a + \gamma\varphi/2) - H\{cK_1 + \varphi(M_4-3)/6\} + cK_q.\end{aligned}$$

Equate this with (10) multiplied by $\sqrt{n}$, then multiply by n and collect terms. This is (8). Under closed-support contact, every term defining B and ξ vanishes. □

Appendix A represents this same E_n by a compensated Fourier integral. The support residual also explains why conservation must not be used without contact. For instance, if X is uniform on $[-\sqrt{3}, \sqrt{3}]$ and $H = 0$, there are no support gaps, yet $B = cK_q/n^{3/2} = 3c\sqrt{3}/(40n^{3/2}) > 0$. The Lebesgue support integral in B is therefore essential.

2.4 A positive resource and its ordered phase price

The geometric inequalities in the next section apply to the same marginal and its actual support gaps. Lemma 3.1 proves that, for $\sigma = \sqrt{10}$,

$$D = (3+\sigma)\rho - \gamma - 3J \geq 0. \tag{13}$$

We use this fact here and give its signwise moment and gap proof once, in Section 3.

Specialize to $c = C_{\mathrm{E}} = (3+\sigma)\phi(0)/6$ and contact. With $w = H^2/(2n)$, $e = e^{-w}$, the identity becomes

$$\mathcal{R}_c = (n+H^2)a + nc\rho(1-e) + \frac{n\phi(0)e}{6}D + n^{3/2}L. \tag{14}$$

If $a \geq 0$, every term is nonnegative. This specialization to C_{E} is essential for this definition of D.

Corollary 2.4 (Raw bound, density, and phase price)**.** *At an attaining pair with $a \geq 0$ and $c = C_{\mathrm{E}}$, assume $n \geq N \geq 2$, $\rho \geq l > 0$, $0 < c_- \leq c$, and a proved raw bound $\mathcal{R}_c \leq T < Nc_-l$. A negative T excludes the pair. Otherwise*

$$\gamma + 3J > 0, \qquad e \geq 1 - T/(Nc_-l) > 0,$$
$$\mathcal{U} := (n+H^2)a + n\phi(0)D/6 + n^{3/2}L \leq \mathcal{R}_c,$$
$$\mathcal{U} \leq \mathcal{R}_c - \tfrac{1}{2}(c_-l - T/N)H^2.$$

No condition $D < 2$ or own-score upper bound is required.

Proof. Set $M_c = c\rho - \phi(0)D/6 = \phi(0)(\gamma+3J)/6$. The general master identity gives

$$eM_c = (1+H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n \geq c\rho - T/n > 0.$$

Thus $M_c > 0$ and, since $D \geq 0$, $M_c \leq c\rho$. Dividing by $c\rho > 0$ gives

$$e \geq 1 - \frac{T}{nc\rho} \geq 1 - \frac{T}{Nc_-l} > 0,$$

where $T \geq 0$ is used in the second inequality. This is the density bound, since $\varphi = \phi(0)e$. Regrouping (14) gives $\mathcal{R}_c = \mathcal{U} + n(1-e)M_c$, proving the unweighted bound. Finally

$$n(1-e)M_c = n(e^w - 1)eM_c \geq \frac{H^2}{2}(c\rho - T/n) \geq \frac{H^2}{2}(c_-l - T/N).$$

This prices one term once. It is a bound on $\mathcal{U}$, not an improved bound on the raw remainder. □

The order is essential at each actual index: first a raw ceiling proves $M_c > 0$ and a density floor; only then may the phase term improve $\mathcal{U}$. A smaller resource ceiling cannot replace the raw ceiling in that license. Common scalar bounds on a closed moment cell make the conclusions uniform even when the maximizing law varies with n.

3 The sign split and the physical gap defect

The sign split gives two conditional means and two centered conditional noises. This is an exact description of the marginal; no nearby Bernoulli law has yet replaced it. The gap defect then measures how much room remains for these noises and for motion of the support endpoints.

3.1 Conditional means and the normalization

For a centered variance-one law with finite third absolute moment, put

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad v = pq, \quad m = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \qquad B_0 = \mathbf{1}_{\{X>0\}}.$$

Both sign classes have positive mass. An atom at zero, if present, belongs to the nonpositive class. Let $V = \mathbb{E}(X \mid B_0)$ and $W = X - V$. Conditional centering and the variance decomposition give

$$V = -m/q \quad (B_0 = 0), \qquad V = m/p \quad (B_0 = 1),$$
$$\mathbb{E}(W \mid B_0) = 0, \qquad \theta = \mathbb{E}W^2 = 1 - m^2/v \in [0, 1).$$

Thus $1 - \theta$ is the variance carried by the two conditional means. Their distance is

$$\ell = m/v = \sqrt{(1-\theta)/v} > 0.$$

Set $U = W/\ell$. Since $V = \ell(B_0 - p)$, we obtain

$$X = \ell(B_0 + U - p), \qquad \mathbb{E}(U \mid B_0) = 0. \tag{15}$$

Writing U_i for the conditional law of U given $B_0 = i$, define

$$w_i = \mathbb{E}U_i^2, \quad \mu_i = \mathbb{E}U_i^3, \quad s = qw_0 + pw_1 = \mathbb{E}U^2 = \theta/\ell^2,$$
$$\lambda = ns, \qquad v + s = \ell^{-2}.$$

In particular $X = (B_0 + U - p)/\sqrt{v+s}$. The noise U is measured in units of the distance ℓ, whereas W is measured in the original units of X. If $s = 0$, both conditional variances vanish and $U_0 = U_1 = 0$ almost surely: this is precisely the standardized Bernoulli case. All third moments here are finite. Higher moments will only be used after support has been acquired.

The signs remain those of the original split. On the nonpositive class, $|X|^3 = \ell^3(p - U_0)^3$; on the positive class it equals $\ell^3(q + U_1)^3$. Expanding, and cancelling the conditional first moments, gives

$$\rho/\ell^3 = v(1 - 2v) + 3v(w_0 + w_1) - q\mu_0 + p\mu_1,$$
$$\gamma/\ell^3 = v(q - p) + 3v(w_1 - w_0) + q\mu_0 + p\mu_1,$$
$$\beta/\ell^2 = v(q - p) + pw_1 - qw_0. \tag{16}$$

For example, $qp^3 + pq^3 = v(1 - 2v)$ and the quadratic terms are $3qpw_0 + 3pqw_1$. All three identities use the same two conditional laws; their moments are not independent choices.

3.2 The moment circle and the actual gaps

The next bounds have no compact-support hypothesis. They will first control a maximizing law coarsely, before a small central gap is known.

Lemma 3.1 (Moment and gap bounds)**.** *For $\sigma = \sqrt{10}$ and the actual J of Section 2,*

$$\gamma^2 + 4m^2 \le \rho^2, \quad J \le \rho + 2m, \quad (2\rho)^{-1} \le m \le \tfrac{1}{2}, \tag{17}$$

$$|\beta| < 1, \qquad \rho \ge \frac{1 + \beta^2}{\sqrt{1 - \beta^2}}, \qquad D := (3 + \sigma)\rho - \gamma - 3J \ge 0. \tag{18}$$

If zero belongs to the closed support, $D \ge (\sigma - 1)\rho$. Otherwise its central gap $(-A, B)$ has $A, B > 0$, $AB \le 1$, and $J \le \rho + 2mAB$.

Proof. Write the actual moments on the two signs as

$$a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2, \qquad b_\pm = \mathbb{E}X_\pm^3 = (\rho \pm \gamma)/2.$$

All four are positive. Cauchy–Schwarz on each sign gives

$$a_\pm^2 \le m b_\pm, \qquad m^2 \le p a_+, \qquad m^2 \le q a_-.$$

The last two inequalities imply $m^2(1/a_+ + 1/a_-) \le p + q = 1$. Since $a_+ + a_- = 1$, this says $m^2 \le a_+ a_-$. Consequently

$$b_+ b_- \ge \frac{(a_+ a_-)^2}{m^2} \ge m^2, \qquad \rho^2 - \gamma^2 = 4 b_+ b_- \ge 4m^2.$$

This is the moment circle. The same inequalities give

$$m\rho \ge a_+^2 + a_-^2 = (1 + \beta^2)/2, \qquad 2m \le \sqrt{1 - \beta^2}.$$

Here $a_+ a_- > 0$, so $|\beta| < 1$ and division is legitimate. Combining these two bounds proves the stated lower bound for ρ in terms of β. Finally $\mathbb{E}|X| = 2m \le 1$, and another Cauchy inequality gives $1 \le \sqrt{\mathbb{E}|X|\mathbb{E}|X|^3} = \sqrt{2m\rho}$. These prove both bounds for m.

For the gap estimate, recall $\tau(t) = \mathbb{E}[X \mathbf{1}_{\{X \ge t\}}]$. At negative t, centering rewrites it as $-\mathbb{E}[X \mathbf{1}_{\{X < t\}}] \ge 0$; at nonnegative t its sign is immediate. Integrating separately over the two half-lines gives

$$\int_{\mathbb{R}} \tau(t) dt = 1, \qquad 2 \int_{\mathbb{R}} |t| \tau(t) dt = \rho. \tag{19}$$

For a positive gap (a, b), $(b - a)^2 \le b^2 - a^2 = 2 \int_a^b t \, dt$; reflection gives the corresponding negative-gap inequality. A central gap $(-A, B)$ has constant weight m and

$$m(A + B)^2 = 2m \int_{-A}^{B} |t| dt + 2mAB.$$

Moreover $(X + A)(X - B) \ge 0$ almost surely, so $0 \le \mathbb{E}[(X + A)(X - B)] = 1 - AB$. Summing the gap inequalities against their constant weights, and using nonnegative Tonelli for countably many gaps, gives $J \le \rho + 2mAB \le \rho + 2m$. If zero is in the support there is no central gap, and the stronger estimate is $J \le \rho$.

Cauchy in $\mathbb{R}^2$, now applied to $(\gamma, 2m)$ and $(1, 3)$, gives

$$\gamma + 6m \le \sqrt{10} \sqrt{\gamma^2 + 4m^2} \le \sigma\rho.$$

Hence the defect is the sum of two nonnegative losses:

$$D = 3(\rho + 2m - J) + (\sigma\rho - \gamma - 6m) \ge 0.$$

In the absence of a central gap, $J \le \rho$ and $\gamma \le \rho$ give $D \ge (\sigma - 1)\rho$. This proves all assertions, including the case of a zero atom. □

3.3 Keeping the central and outer costs together

Suppose the central gap $(-A, B)$ has been acquired. Define

$$T_+ = \mathbb{E}X_+^3 - mB^2, \qquad T_- = \mathbb{E}X_-^3 - mA^2,$$
$$R_+ = \mathbb{E}X_+^2 - mB, \qquad R_- = \mathbb{E}X_-^2 - mA.$$

These quantities measure excess beyond the two gap endpoints. They are nonnegative because every positive support point is at least B, and every negative magnitude is at least A. Let $J_o = J - m(A+B)^2 = J_o^+ + J_o^-$ be the outer-gap contribution. A positive outer gap has $a \geq B$, so

$$(b-a)^2 \leq 2\int_a^b (t-B)dt.$$

Multiplying by its constant gap weight and summing gives

$$\begin{aligned} J_o^+ &\leq 2\int_B^\infty (t-B)\tau(t)dt \\ &= \mathbb{E}[X_+(X_+-B)^2] = T_+ - 2BR_+ \leq T_+. \end{aligned}$$

The negative-side calculation is the same in magnitudes: $J_o^- \leq T_- - 2AR_- \leq T_-$. Endpoint atoms do not change these Lebesgue integrals; their mass is already included in τ.

Substitute $\rho = m(A^2+B^2)+T_++T_-$, $\gamma = m(B^2-A^2)+T_+-T_-$ and $J = m(A+B)^2+J_o$ into the definition of D. Using $\sqrt{\sigma+1}\sqrt{\sigma-1} = 3$ completes the central square:

$$\begin{aligned} D = m\{\sqrt{\sigma+1}A - \sqrt{\sigma-1}B\}^2 + (\sigma-1)T_+ + (\sigma+1)T_- \\ + 3(T_+ + T_- - J_o). \end{aligned} \tag{20}$$

Every displayed term is nonnegative. In particular $T_+ + T_- \leq D/(\sigma-1) \leq D/2$. If $D < 2$, then

$$m(A^2+B^2) = \rho - T_+ - T_- \geq 1 - D/2.$$

Since $m \leq 1/2$ and $A, B > 0$, this gives $A + B \geq \sqrt{2-D}$. The signed entry estimates retain the full central square rather than using only this weaker distance bound.

For direct acquisition it is useful to retain the two sharper outer-gap bounds before dropping $R_\pm$. They give

$$D \geq (\sigma-1)\mathbb{E}X_+^3 + (\sigma+1)\mathbb{E}X_-^3 - 6mAB + 6BR_+ + 6AR_-.$$

Replace J_o by $T_+ + T_- - 2BR_+ - 2AR_-$ in the preceding expansion. Direct acquisition substitutes the conditional means and noises into this single budget, retaining their joint cost.

4 Coarse entry from contact

Fix an attained pair with $a \geq 0$, $n \geq N = 42$, and true predecessor $C_{n-1} \leq c = C_{\mathrm{E}}$. All bounds below concern this one marginal and raw threshold. They pass through two quadratic and four signed updates; their finite coverage conditions remain explicit.

There are three tasks, and their order matters. Contact first turns the predecessor bound into a support radius without using higher moments. The radius then gives moment ceilings and a Fourier error bound. Finally, two coarse passes acquire a central gap, after which four signed passes retain its additional information. The finite conditions record the resulting inequalities on complete closed covers; they do not replace the analytical implication from the actual law to each cover. Throughout, $N = 42$ is the evaluation floor; the witness may have any actual $n \geq N$.

4.1 Support before higher moments

Theorem 4.1 (Initial entry)**.** *Assume Shevtsova's external structural inequality [5]*

$$\sqrt{n}\Delta_n(P) \le \frac{3031}{10000}(\rho + 323/500). \tag{21}$$

Then $\rho < 7361/4000$, $|\beta| < 321/500$ *and* $\operatorname{supp} P \subset (-96/25, 96/25)$, *uniformly in* H.

Proof. Since $c\rho \le \sqrt{n}\Delta_n(P)$ and $c > 819/2000$, division gives

$$\rho < \frac{(3031/10000)(323/500)}{819/2000 - 3031/10000} = \frac{7361}{4000}.$$

Signwise Cauchy gives $\rho \ge (1+\beta^2)/\sqrt{1-\beta^2}$, increasing in $|\beta|$. At $r = 1841/1000$, $b = 321/500$,

$$(1+b^2)^2 - r^2(1-b^2) = \frac{93288849}{50000000000} > 0.$$

Put $v = n-1$, $s = \sqrt{n/v}$, $F_v(x) = \Phi(x/\sqrt{v})$ and $e_n = n(s-1) - 1/2$. Contact and the predecessor imply

$$0 \le H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} - na \quad (t \in \operatorname{supp} P), \tag{22}$$

where

$$H_n/n^{3/2} = F_v(H-t) - F_{v+1}(H) + tF'_{v+1}(H) - (t^2-1)F''_{v+1}(H)/2.$$

The extra $-na$ is nonpositive; no unknown-score comparison is used. The predecessor was applied to the same marginal and shifted raw event:

$$\mathbb{P}(S_{n-1} \le H - t) \le F_v(H-t) + c\rho/\sqrt{v}.$$

Combining this with contact and $\sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} = c\rho + a$ gives the displayed inequality. Thus the comparison uses the known fixed target c, not the unknown attained score $c + a/\rho$. Both events are closed, including at atoms of the sum laws. Rationalization gives $ne_n = s^2(2s+1)/(2(s+1)^2) < 793881/2073680 < 48/125$ and $s < S := 81/80$, hence $e_n < \eta := 8/875$; the rationalized expression increases for $s \ge 1$.

The heat identity $\partial_v F_v = F''_v/2$, spatial Taylor expansion to order four and variance expansion to order two give, for $u \ge 1$,

$$H_n(H, \pm u)/s^3 = \pm A(1-z_v^2)\phi(z_v) + B\phi'''(z_v) + \mathcal{E}, \tag{23}$$

where $z_v = H/\sqrt{v}$, $A = (u^3 - 3u)/6$, $H_4 = u^4 - 6u^2 + 3$, $B = H_4/(24\sqrt{v})$ and

$$|\mathcal{E}| < E(u) := u^5/4100 + 3u/820 + 17(3u^2-2)/78720.$$

The four remainders after division by s^3 are bounded by

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120v}, \quad \frac{\|\phi^{(5)}\|_\infty}{48v^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v}, \quad \frac{|u^2-1|\|\phi^{(5)}\|_\infty}{16v^{3/2}}.$$

Gaussian inversion bounds these derivative norms by $6/5$ and $51/20$; use $v \ge 41$, $\sqrt{v} > 6$. No higher moment of P enters. The derivative bounds can be taken as $3\phi(0) < 6/5$ and $8/\pi < 51/20$. These errors are uniform in H and concern only the Gaussian comparison. The support proof therefore has not assumed any of the fourth- or fifth-moment ceilings

which will follow from it. For $A, B > 0$, $B \le A/4$, the leading expression is at most $\phi(0)(A + 9B^2/(4A))$ for both signs: for the positive sign on $|z_v| \le 1$, complete the square in $A - Az_v^2 + 3B|z_v|$; outside it the first term is nonpositive. For the negative sign use $2e^{-3/2}A < A/2$ and $\|\phi'''\|_\infty < 2\phi(0)$. The latter follows from $3x - x^3 \le 2$ on $[0, \sqrt{3}]$ and $x^3 e^{-x^2/2} \le 3\sqrt{3}e^{-3/2} < 2$ beyond it.

Freeze the preceding r, b, and set

$$Q(u) = -u^3 + 3ru^2/2 + 3bu + \eta r,$$

$$P(u) = \tfrac{2}{5}S^3\{A^2 + H_4^2/(256 \cdot 41)\} + S^3 E(u)A + (819/2000)Q(u)A.$$

At $R = 96/25$ we have $A > 0$, $0 < H_4 < 36A$, $Q(R) < 0$ and

$$-P(R) = \frac{2214258283536204102723}{10250000000000000000000} > \frac{1}{5}.$$

The sign $Q(R) < 0$ is checked before replacing c by its lower bound. Since $A > 0$, the upper bound in (22), multiplied by A, is strictly below $P(R)$. A boundary test alone would still allow a more distant support point, so the derivative signs below are needed. Contact is therefore impossible at either boundary. For the whole rays,

$$|\partial_t H_n(H, t)| < \tfrac{2}{5}(t^2/2 + 1/2 + e_n).$$

Indeed, with $y = (H - t)/\sqrt{n}$, $z = H/\sqrt{n}$, the derivative is $n\phi(z) + Ht\phi(z) - ns\phi(sy)$. The spatial second-order error costs $\phi(0)t^2/2$, while $\partial_s[s\phi(sy)] = -\phi''(sy)$ costs $\phi(0)n(s-1)$. Put $L(u) = \frac{2}{5}(u^2/2 + 1/2 + \eta) + (819/2000)Q'(u)$. At R, all Q, Q', Q'', L, L', L'' are negative. As $Q''' = -6$ and $L'' = 2/5 - 6(819/2000) < 0$, the signs persist outwards, so both actual radial upper expressions decrease from negative boundary values. □

The same heat/ray argument licenses a later radius in $\{2.50, 2.51, \dots, 3.84\}$: freeze the cell bounds r, b, then check $A > 0$, $0 < H_4 < 36A$, $Q(R), P(R) < 0$ and the six ray signs before using that radius in any higher-moment bound. A failed test leaves the preceding radius in force. Strict gates exclude boundary atoms as well as the exterior rays, so a subsequent closed moment box may use the selected radius as a non-strict support ceiling.

4.2 The common Fourier input

After acquiring support, put $M_j = \mathbb{E}|X|^j$. The measure $d\mu_0 = (x-1)^2(x+1/2)dP$, $x = |X|$, has mass $\rho - 1$, and

$$M_4 - 1 = \int \frac{(x+1)^2}{x+1/2} d\mu_0, \qquad M_5 - 1 = \int \frac{x^3 + 2x^2 + 3x + 3/2}{x + 1/2} d\mu_0.$$

The ratios increase on $x \ge 0$; write k_4, k_5 for their values at R. Indeed they are $x + 3/2 + 1/[4(x + 1/2)]$ and $x^2 + 3x/2 + 9/4 + 3/[8(x + 1/2)]$, with nonnegative derivatives. Their endpoint values bound the integrals against the nonnegative measure μ_0; this is the reason the ceilings improve as ρ tends to one. Consequently

$$M_4 \le \min\{R\rho, 1 + k_4(\rho - 1)\}, \quad M_5 \le \min\{RM_4, 1 + k_5(\rho - 1)\}, \quad M_6 \le RM_5. \tag{24}$$

Also $\gamma^2 \le M_4 - 1$ by $\mathbb{E}(X^2 - 1 - \gamma X)^2 \ge 0$. For two iid copies, $\mathcal{E} = \{(X - X')^2 \ge 9/25\}$ satisfies

$$\mathbb{P}(\mathcal{E}) \ge \alpha := \frac{(2 - 9/25)^2}{2M + 6}, \quad |X - X'| \ge 3/5 \text{ on } \mathcal{E}, \quad \mathbb{E}[(X - X')^2 \mid \mathcal{E}] \le 2/\alpha. \tag{25}$$

Here the event has been obtained without assuming a central gap. Its mass and conditional moment refer to two independent copies of the actual marginal. For $M \geq M_4$, Cauchy proves the mass bound from $\mathbb{E}[(X-X')^2;\mathcal{E}] \geq 2-9/25$ and $\mathbb{E}(X-X')^4 = 2M_4+6$; the last bound uses $\mathbb{E}(X-X')^2 = 2$.

Appendix A proves the complete compensated Fourier bound, including atoms and the comparator tail. Its input is an acquired radius R, ceilings $M \leq 11.1$, $g \leq 1.85$ for $M_4, |\gamma|$, a sixth-moment ceiling, and $|(3-M_4)/24| \leq A_4$, $|(10\gamma - \mathbb{E}X^5)/120| \leq Z_5$. Use the complete LOW triple L_0, L_H, L_2 of Proposition A.2, with the sharper fifth-order coefficient proved there. An actual event of mass at least α, separation at least d_E and conditional second moment at most M_E supplies the six-term HIGH bound K_H of Proposition A.3, under all its damping and all-index gates. Insert each event option into all six terms before minimization. The output is

$$|E_n| \leq K_F + s_F|H|, \qquad K_F = \tfrac{16}{25}(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = \tfrac{16}{25}L_H. \tag{26}$$

It is uniform in actual $n \geq N$, even when the law and threshold vary. The LOW term uses moments of this marginal; the HIGH term uses an event under two independent copies of it. A moment rectangle does not itself supply a characteristic function or a separation event. When a later pass improves a moment or acquires a second event, all the HIGH terms must use that option together. The minimum of the complete estimates still bounds the same E_n.

4.3 Coarse coefficients and two passes

The constant coefficient of c in (7) is

$$A^{\mathrm{law}} = 3\beta(J+\gamma)/2 + 3(\rho J_1 - J_2) + \rho M_4/2 - M_5/4 - \rho/2.$$

This is the resource contribution independent of H, before its Fourier error is added. The linear term retains K_1 and Ω; the quadratic term will supply a raw ceiling. We start with bounds that do not assume a central gap:

$$-M_4/6 \leq J_1 \leq M_4/2, \qquad -3J_2 - M_5/4 \leq 0, \qquad J \leq \rho + 1. \tag{27}$$

For a gap (u,v) of length h, the first pair follows from $\int (t-u)t + \frac{1}{2}\int t^2 = hv^2/2$ and $\frac{3}{2}\int t^2 - \int (t-u)t = h(2u+v)^2/6$; sum against its weight and use $\int t^2\tau = M_4/3$. On negative gaps reflect and use $\frac{1}{3}\int_u^v x^3dx - \int_u^v (v-x)x^2dx = u^3(v-u)/3 \geq 0$ for the second bound; positive pieces improve it. The last uses $J \leq \rho + 2m$. For $1 \leq r \leq \rho$, $R \leq 6$, choose

$$\Gamma(R,r) \geq \max_{r\leq x\leq R} \left\{ -x^2/4 + (R+r)x/3 - Rr/2 + r^3(2R-r)/(12x^2) \right\}_+ . \tag{28}$$

Then

$$3(\rho J_1 - J_2) \leq 3\Gamma(R,r)(\rho - \gamma)/2 + 3\rho^2 J/8. \tag{29}$$

On a negative gap discard $|t| < \rho$, whose charge is nonpositive. Extend the rest to $\int_r^v (R-x)x(x-r)dx$; dividing by v^2 gives (28). Tonelli charges its bound against $\mathbb{E}X_-^3 = (\rho-\gamma)/2$. On positive pieces use $t(\rho - t) \leq \rho^2/4$ and $\sum \tau \int (t-u) = J/2$. To enclose Γ, split $[r,R]$ into 64 closed intervals, maximize the quadratic at its clipped vertex $2(R+r)/3$ and the positive reciprocal term at the left endpoint. Alternatively $\Gamma \leq 5/2$: at $r=1, R=6$ this is $3x^4 - 28x^3 + 66x^2 - 11 > 0$ on $[1,6]$, whose endpoint values are $30, 205$ and interior critical values $(85 \mp 35\sqrt{5})/2 > 0$. The integral increases with R and decreases with r.

On $\rho \in [l,u]$, $|\beta| \le b$, $M_4 \le M$, $|\gamma| \le g$ and $0 \le \rho - \gamma \le \Delta$, the square $V = M_5 - 2\rho M_4 + \rho^3 \ge 0$ gives two complete constant bounds:

$$A_1(\Delta) = 3b(2u+1)/2 + 3\Gamma(R,l)\Delta/2 + 3u^2(u+1)/8 + u^3/4 - l/2,$$
$$A_2 = 3b(2u+1)/2 + 2uM - l/2. \tag{30}$$

For $K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2$, intersect the applicable bounds

$$K_{\mathrm{rad}} = 3ub + \frac{R(R+2-3l/2)}{R+1/2}(u-1), \quad K_{\mathrm{univ}} = 6u - 3u^2/2 + 3b,$$
$$K_R = Ru/2 + \max_{r=l,u} |R/2 - 3r/2|g + 3b.$$

The first follows from $K_1 + 3\rho\beta = \int \mathrm{sgn}(X)x(x+2-3\rho/2)/(x+1/2)d\mu_0$. This ratio is convex in x, decreases in ρ, and its negative part is at most $31/40$ for $\rho \le 1.85$; require the endpoint ceiling to exceed that number. The second uses $|\mathbb{E}[X|X|^3] - 3\gamma| \le 3\rho$ for $R \le 6$, $\rho \le 2$; the third uses $0 \le \mathbb{E}X_\pm^4 \le R(\rho \pm \gamma)/2$. Finally $|\varphi\Omega| \le v_H := \frac{2}{5}\max\{1/2, 2M/3 - 1/2\}$ for $\Omega = J_1 + (M_4 - 3)/6$.

Lemma 4.2 (Signed fourth-moment charge)**.** *Put $h_* = 1 + b$, $a(r) = (3r - R)/2$. Another bound for $|K_1|$ is $\max_{r\in[l,u]}\{Rr - 3r^2/2 + 3b + (1+b)a(r)^2/2\}$. If*

$$h_* \ge a(u), \quad 3 + 2a(l)h_* - 3h_*^2/2 \ge 0, \quad R - 3u + 3h_*^2/2 \ge 0,$$

intersect it with $Ru - 3u^2/2 - 3 + 3h_* + a(u)h_*^2 - h_*^3/2$.

Proof. For $\epsilon = \pm 1$, put $h = 1 - \epsilon\beta > 0$, $\delta = \rho - \epsilon\gamma$. Signwise Cauchy and $m \le 1/2$ give $\delta \ge h^2$; fourth moments give

$$\epsilon K_1 \le R\rho - 3\rho^2/2 - 3 + 3h + a(\rho)\delta - \delta^2/(2h).$$

Unrestricted square completion proves the first bound. Under $\delta \ge h^2$, the last three terms maximize to $h(3 + a^2/2)$ for $h \le a$, and $3h + ah^2 - h^3/2$ for $h \ge a$. They agree at a. The first is increasing; the second has derivative $3 + 2ah - 3h^2/2$, nonnegative on $[0, h_*]$ by the second gate. The first gate selects its branch and the third makes its remaining ρ derivative nonnegative. Evaluate the first quadratic at endpoints and its eligible concave vertex. □

Call the applicable minimum K; no sign of $\beta\gamma$ is assumed. Each candidate bounds the whole K_1. The conditional candidate is used only where all three derivative gates hold; otherwise the unconditional ones remain available. Reflection in the proof establishes the two sign inequalities without reorienting the actual marginal during acquisition.

Use 24 closed cells $[r_i, r_{i+1}]$, $r_i = 1 + 3361i/96000$, and

$$c_- = .4097321837023 < c < c_+ = .4097321837025,$$
$$\phi_l = 997/2500 < \phi(0), \qquad \sigma_l = 3.162277660 < \sqrt{10}.$$

The alternating Machin bounds and rational squares of Lemma B.1 license these constants. Put $\sigma_u = 3.162277661 > \sqrt{10}$.

The first coarse pass.

This pass needs neither a positive density floor nor a gap assumption. It first seeks a raw ceiling below Nc_-l; conservation can then give a smaller resource ceiling and a defect bound. In pass 1 take $A = \min(A_1(2u), A_2)$, $f(t) = tK + v_H + s_F$, $q(t) = tl - g/5 > 0$. With overlines denoting upward rounding to 10^{-10}, set

$$\begin{aligned} T_1 &= \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/(4q(t))\}}, \\ \eta_1 &= (c_-l - T_1/N)/2, \quad U_1 = \min\{T_1, \overline{\max_{t=c_-,c_+} [K_F + tA + f(t)^2/(4(q(t)+\eta_1))]}\}, \\ d_1 &= \overline{6U_1/(N\phi_l)}. \end{aligned} \tag{31}$$

Check $0 \le T_1 < Nc_-l$ before using η_1. The raw quadratic is affine in c at each $|H|$, so maximize at endpoints before completing squares. Corollary 2.4 then gives T_1, U_1, Nd_1. The two ceilings have different uses:

$$\begin{gathered} \mathcal{R}_c \le T_1 < Nc_-l \implies \varphi \ge \phi_l(1 - T_1/(Nc_-l)), \\ nD \le 6U_1/\phi_l \le Nd_1. \end{gathered}$$

The first licenses the density floor, while the second controls the nonnegative resource $\mathcal{U}$. A smaller U_1 cannot replace T_1 in its Gaussian license merely because it is a better resource bound.

The second coarse pass.

The first defect ceiling now constrains the signed third moment of the same law. This feedback improves both the constant charge and the quadratic curvature before the next ordered update. For pass 2 the moment circle gives $D \ge \sigma\rho - \gamma - 3\sqrt{\rho^2 - \gamma^2}$. Put $a_0 = \sigma_l - d_1/l$ and

$$y_l \le Y(a_0) := \begin{cases} -1, & a_0 \le -1, \\ (a_0 - 3\sqrt{10 - a_0^2})/10, & -1 < a_0 \le \sqrt{10}. \end{cases} \tag{32}$$

This smaller root inverts the increasing branch of $y + 3\sqrt{1-y^2}$ on $[-1, 1/\sqrt{10}]$; $a_0 - y \ge 0$ prevents the false squared branch. Round the root downward. Thus

$$\begin{gathered} \gamma \ge g_l := \max\{-g, \sigma_l l - 3 - d_1, \min(ly_l, uy_l)\}, \\ \Delta = \min\{2u, (1-\sigma_l)l + 3 + d_1, u - g_l\}, \quad \phi_- = \phi_l(1 - T_1/(Nc_-l)), \\ A = \min\{A_1(2u), A_1(\Delta), A_2\}, \quad q(t) = \max\{tl - g/5, tl + \tfrac{1}{2}\min(\phi_- g_l, (2/5)g_l)\}. \end{gathered}$$

Here $2m \le 1$ gives the additional bounds and both alternatives for q have slope l. Repeat the raw step, intersect with T_1, price phase only then, and intersect the resource with U_1 and d_2 with d_1.

Certificate condition 4.3 (Two coarse passes)**.** All coefficient, support, Fourier, denominator and strict raw gates above hold on all 24 cells, with outward ceilings

	$\max T$	$\max U$	$\max d$	strict gap gates
1	13.2818901598	10.9330610254	3.9164138937	23
2	11.5164967977	9.9790315907	3.5746638454	24

and $d_2 < (\sigma_l - 1)l$ in every second-pass cell, with minimum margin

$$\frac{526024602543}{1600000000000} > 0.$$

This is a finite exact obligation. Zero in the support would give $J \le \rho$ and $D \ge (\sigma-1)\rho$, contradicting its gap test. Centering therefore acquires an actual central gap $(-A, B)$, $A, B > 0$. This is the logical transition recorded by the table: only 23 first-pass cells satisfy the strict gap test, so a uniform gap cannot be used there. The second pass satisfies it in every cell. Only now may we use gap endpoints, the opposite-sign event, and separate central and outer costs.

4.4 Signed refinements of one physical law

We now retain the signs of β and γ, the common first moment m, and the two sign classes of one marginal. The boxes below give necessary enclosures; they do not assert that independently chosen coordinates are realizable moments.

Common moments and support constraints.

Put $a_\pm = (1 \pm \beta)/2$, $b_\pm = (\rho \pm \gamma)/2$. Signwise Cauchy gives

$$m^2 \le a_+ a_-, \quad m b_\pm \ge a_\pm^2, \quad m \ge \max\{(1+\beta^2)/(2\rho), a_+^2/b_+, a_-^2/b_-\},$$
$$\frac{(1+\beta)^{3/2}}{\sqrt{1-\beta}} - \rho \le \gamma \le \rho - \frac{(1-\beta)^{3/2}}{\sqrt{1+\beta}}.$$

Both signs have positive mass and first moment, hence $|\beta| < 1$. Projection of $X|X|$ onto $(X, |X|)$, whose Gram matrix is $\left(\begin{smallmatrix} 1 & \beta \\ \beta & 1 \end{smallmatrix}\right)$, gives $M_4 \ge \rho^2 + (\gamma - \rho\beta)^2/(1-\beta^2)$. Equivalently, minimize $\mathbb{E}(X|X| - uX - v|X|)^2$ over real u, v. The Gram matrix is positive definite because $|\beta| < 1$, and completing the square gives this lower bound. It couples the signed moments before any interval endpoints are substituted. Integrating $(1, |X|)^{\mathsf{T}}(1, |X|)$ against $(R - |X|)dP$, $|X|(R-|X|)dP$ and $|X|^2(R-|X|)dP$ on each sign gives positive-semidefinite matrices, whose determinants imply

$$2m + \sum_\pm \frac{(Rm - a_\pm)^2}{D_\pm} \le R, \quad M_4 \le R\rho - \sum_\pm \frac{D_\pm^2}{Rm - a_\pm}, \quad D_\pm = Ra_\pm - b_\pm,$$
$$\mathbb{E}X_\pm^5 \le R\mathbb{E}X_\pm^4 - \frac{(Rb_\pm - \mathbb{E}X_\pm^4)^2}{D_\pm}.$$

Use corrections only at positive denominators. A positive numerator with zero denominator excludes only by its original determinant inequality; zero-over-zero corrections are dropped. The same m is retained on both signs. For the first inequality the two determinants are summed, so that the total mass is bounded by one while sharing exactly that m. Independent first moments on the two signs would not give the same constraint. Integrating $(x-a)^2(R-x) \ge 0$ gives also $2m \le (R - \rho + 2a + Ra^2)/(a(2R+a))$; use $a = 2/10, \ldots, 12/10$.

Three bounds for one fifth-moment loss.

The coarse constant used the square

$$V = M_5 - 2\rho M_4 + \rho^3 = \mathbb{E}\{|X|(|X|^2 - \rho|X|)^2\},$$
$$\rho M_4/2 - M_5/4 = \rho^3/4 - V/4.$$

Signed information gives three different lower bounds for this same $V/4$. For $\delta = \gamma - \rho\beta$, they improve $\rho M_4/2 - M_5/4 \le \rho^3/4$:

$$P_1^* = \frac{(\rho+\gamma)\delta_+^2}{8(1+\beta)^2} + \frac{(\rho-\gamma)(-\delta)_+^2}{8(1-\beta)^2}, \quad P_2^* = \frac{\delta^4}{4\rho(1-\beta^2)^2}, \quad P_3^* = \frac{\delta^2}{8m}.$$

For the first, signwise Cauchy gives $c_4 \geq b_3^2/a_2$, $f_5 \geq c_4^2/b_3$, so $\rho c_4/2 - f_5/4 \leq \rho^2 b_3/4 - (c_4 - \rho b_3)^2/(4b_3)$; minimize the square over that half-line. The second follows from $M_4^2 \leq \rho M_5$ and the projection bound. For the third, $V = \mathbb{E}[|X|(|X|^2 - \rho|X|)^2]$ and weighted Cauchy give $\delta^2 \leq 2mV$. Subtract their maximum, never their sum. Here $\delta = \mathbb{E}[|X| \operatorname{sgn}(X)(|X|^2 - \rho|X|)]$, which explains the weighted Cauchy step. Each deduction bounds the entire loss; adding the three would count it repeatedly. A zero atom has zero weight in these identities.

Endpoint information from the physical defect.

A central gap gives an event distance even when the current defect ceiling is not below two. The physical decomposition (20) gives $T_+ + T_- \leq D/(\sigma - 1)$ for $T_+ = b_+ - mB^2$, $T_- = b_- - mA^2$. Also $J \leq \rho + 2mAB$ gives $6mAB \geq (\sigma - 1)\rho - D$. On $\rho \geq l$, $D \leq d < (\sigma_l - 1)l$, $m \leq m_u$,

$$(A + B)^2 \geq \{l - d/(\sigma_l - 1) + ((\sigma_l - 1)l - d)/3\}/m_u > 0. \tag{33}$$

Retain $A \leq a_-/m$, $B \leq a_+/m$ and $D \geq \sigma\rho - \gamma - 6m$. For the distance bound, use $m(A^2 + B^2) = \rho - T_+ - T_-$ and add the lower bound for $2mAB$ before dividing by $m \leq m_u$. Both the third-moment mass and the endpoint product are retained. In particular

$$m \geq m_* := \max\{0, (\sigma_l l - \gamma_+ - d)/6\}, \qquad |\beta| \leq \sqrt{1 - 4m_*^2}. \tag{34}$$

An $m_* > 1/2$ excludes the cell. Intersect beta afresh in each gamma slice before recomputing dependent moments. A contraction obtained in one gamma slice cannot be carried into the next: it may exclude a different feasible beta value. Each slice therefore starts from the parent beta interval. The stronger outer-gap inequality gives $D \geq (\sigma - 1)b_+ + 6a_+B - (\sigma + 5)mB^2$. If $D < (\sigma - 1)b_+$, its concave quadratic forces B above the positive root; the analogue uses $A, a_-, b_-, \sigma + 1, \sigma + 7$. Round positive coefficients down, negative coefficients up and roots down.

The actual opposite-sign event.

There is no atom at zero after gap acquisition. For independent copies, $\{XX' < 0\}$ has probability $2pq$ and separation at least $A + B$. This improves the universal event without inserting a binary surrogate for the actual law. Cauchy gives $p \in [2m_-^2/(1+\beta_+), 1-2m_-^2/(1-\beta_-)]$ and $2pq \geq (1 + \beta^2)^2/(2\rho^2)$. Exactly,

$$\mathbb{E}[(X - X')^2 \mid XX' < 0] = \frac{1 + 4m^2 + \beta}{2p} + \frac{1 + 4m^2 - \beta}{2q}.$$

This increases with m, is affine in β and convex in p, with positive numerators. Use its corner maximum or $2/\alpha$, whichever is smaller, with $\alpha = \max\{(1+\min|\beta|^2)^2/(2u^2), 2\min_{p=p_-,p_+} pq\}$. The downward root of (33) then licenses another complete HIGH option, minimized against the universal event.

Central and outer integral costs.

The central gap can be integrated exactly. The outer gaps consume only the remaining weighted defect budget. For the signed integral charge put

$$P_1 = B^3/3 + AB^2/2 - A^3/6, \quad P_2 = B^4/4 + AB^3/3 - A^4/12, \quad G = \rho P_1 - P_2.$$

The central terms are $J_{1,c} = mP_1$, $J_{2,c} = mP_2$. Since $\partial_B G = B(A + B)(\rho - B)$,

$$G \leq G(A, \rho; \rho) = \rho^4/12 + \rho^3 A/6 - \rho A^3/6 + A^4/12.$$

It increases with A (derivative $(\rho-A)^2(\rho+2A)/6$) and is convex in ρ, so use A_u and both rho endpoints. Positive outer gaps satisfy $(t-a)(\rho-t)\le(\rho-B)_+^2/4$; negative outer gaps cost at most $(R-\rho)^2/4$ after extracting $|t|$. Integration gives

$$3(\rho J_1-J_2)_o\le 3\{(\rho-B)_+^2T_+ + (R-\rho)^2T_-\}/8.$$

The joint budget is $D-mQ$, where $Q=(\sqrt{\sigma+1}A-\sqrt{\sigma-1}B)^2$; intersect it with $T_+\le b_{+,u}-m_lB_l^2$, $T_-\le b_{-,u}-m_lA_l^2$. For Ω use $-RT_-/2\le J_{1,o}\le RT_+/2$. The minimum of P_1 is at B_l and an A endpoint; the maximum is at B_u and A clipped to B_u. Multiply a nonnegative polynomial bound by m_u for an upper bound and m_l for a lower bound, reversing the choices for a negative bound. In particular a negative upper bound for P_1 must be multiplied by the lower m endpoint; using m_u would make it smaller and could discard the actual law. Fourth-semimoment intervals similarly enclose K_1. Retain the full signed pair (K_1,Ω).

4.5 Ordered update and closed coverage

Each cell carries its radius, moments, raw ceiling T and weighted invariant $nD\le Nd$. Intersect all preceding necessary conditions outwardly; monotone bisection for m retains its upper endpoint. Since $n\ge N$, the weighted invariant gives $D\le d$ for all moment and endpoint guards. Its factor n is preserved in the new resource output rather than being replaced by the evaluation floor. Take

$$J_-=\max\{0,((3+\sigma_l)l-\gamma_+-d)/3\},\quad J_+=\min\{u+2m_+,((3+\sigma_u)u-\gamma_-)/3\}.$$

Let B_* be the corner maximum of $3\beta(J+\gamma)/2$, G_* the smaller coarse/central-plus-outer charge and P_* a lower bound for $\max(P_1^*,P_2^*,P_3^*)$. Then

$$A=\min\{B_*+G_*+u^3/4-l/2-P_*,B_*+2uM-l/2\} \tag{35}$$

encloses the complete constant. Let $[k_-,k_+]$, $[\omega_-,\omega_+]$ enclose K_1,Ω. Under $0\le T<Nc_-l$, put $\phi_-=\phi_l(1-T/(Nc_-l))>0$ and

$$\begin{aligned}B_t&=K_F+tA,\quad F_t=\max_{k=k_\pm,\,\omega=\omega_\pm,\,\psi=\phi_-,2/5}|tk+\psi\omega|+s_F,\\ Q_t&=tl+\tfrac12\min\{\phi_-\gamma_-,(2/5)\gamma_-\}.\end{aligned} \tag{36}$$

Thus $\mathcal{R}_c\le B_c+F_c|H|-Q_cH^2$. Successive affine maximization licenses the corners; convexity in c then licenses c_-,c_+ even when $A<0$. Negative γ_- uses the upper density endpoint. Here A bounds the whole constant charge, F_t the whole linear charge, and Q_t the positive curvature from below. Taking corners first provides a valid quadratic for every real H. Its maximum is then found by square completion; no phase grid is involved.

Lemma 4.4 (Ordered step)**.** *If $Q_t>0$ at both endpoints, with all ceilings rounded upwards, set*

$$\begin{aligned}T'&=\min\{T,\max_{t=c_-,c_+}(B_t+F_t^2/(4Q_t))\},\quad \eta=(c_-l-T'/N)/2,\\ U'&=\max_{t=c_-,c_+}(B_t+F_t^2/[4(Q_t+\eta)]),\quad d'=\min\{d,6U'/(N\phi_l)\}.\end{aligned}$$

Negative T' or U' excludes the cell. Otherwise $\mathcal{R}_c\le T'$, $\mathcal{U}\le U'$ and $nD\le Nd'$, with

$$|H|\le\max_{t=c_-,c_+}\frac{F_t+\sqrt{F_t^2+4(Q_t+\eta)\max(0,B_t)}}{2(Q_t+\eta)}. \tag{37}$$

Proof. Complete the raw squares first. Since $T' \le T < Nc_l$, phase pricing gives $\eta > 0$; only now complete the resource squares. Phase pricing increases the curvature from Q_t to $Q_t + \eta$ and can lower its maximum without changing the already proved raw license. The nonnegative resource contains $n\phi(0)D/6$, proving the weighted bound at actual n. A phase where both resource quadratics are negative is impossible; replacing B_t by its positive part enlarges the positive root and gives (37). Intersecting valid bounds preserves the claims; a resource ceiling cannot replace the raw ceiling in its license. □

First license each rho cell's radius by the heat/ray test. Each signed pass splits beta into 8 closed intervals, each necessary gamma range into 8, and the common m range into 2. Restore the original beta interval at every gamma slice before (34). A contradictory necessary condition excludes all descendants of that parent; active plus excluded equals 128 per processed rho cell. Pool upper bounds by maxima and lower bounds by minima; remove a rho cell only when every descendant is excluded. Closed seams may duplicate work, never lose a witness.

These are logical tests on enclosures. Contradictory necessary bounds exclude every actual law in that box and hence all its descendants. Keeping an active box does not assert moment realizability; unnecessary boxes only weaken the estimate. The active/excluded census ensures that no child has disappeared between these alternatives. For export retain all 100 closed panels $[j/100, (j+1)/100]$; a boundary law belongs to both incident panels. Use $\theta = 1 - m^2/(pq)$ with lower m and the full maximum of pq on each cell-panel intersection, including $1/4$ across $1/2$. Pool all bounds from those same cells. Each panel thus gives an implication for every actual law with p in it. Its probability cannot be chosen independently after the moment calculation. These internal panels remain part of the selected finite condition even though the downstream argument uses only a few uniform outputs; dropping their incidences would change that condition.

Certificate condition 4.5 (Four signed passes)**.** Starting from Condition 4.3, all support, moment, actual-event, Fourier, denominator, signed-curvature and raw-license gates hold on this complete four-pass cover. Its 11264 nominal boxes and complete 100-panel export give, on every nonempty panel,

$$\mathcal{R}_c \le 2.1995174, \quad nD \le 32.20665378, \quad D \le .76682509,$$
$$\rho \le 71527/48000, \quad |\beta| \le .50549963, \quad |H| \le 1.83605680, \quad \operatorname{supp} P \subset (-157/50, 157/50).$$

Fourteen rho cells survive and 30 probability panels are nonempty.

Theorem 4.6 (Selected coarse entry)**.** *Under the two finite conditions, every attained pair with $a \ge 0$, $n \ge 42$ and true $C_{n-1} \le C_{\mathrm{E}}$ satisfies these bounds and the complete probability cover, assuming* (21)*.*

Proof. Support licenses higher moments, and the actual events and Fourier appendix bound the full integral. The two coarse passes acquire the gap. Every signed cell applies only necessary conditions of that same law and the ordered step. Induction over all four passes preserves the weighted invariants; closed coverage and outward pooling place every witness in an incident output panel. □

The numerical conditions remain exact finite obligations. The one-strip argument removes only downstream panels; this internal cover stays intact.

5 One acquired strip

The entry calculation will be used only through its weighted defect and raw-threshold bounds. The former already forces the sign split close to Esseen's two-point law. Two applications of contact then bound the entire conditional support, without a further partition of the sign probability. Throughout this section set

$$\sigma = \sqrt{10}, \qquad p_* = (4-\sigma)/2, \qquad B_* = 32.20665378.$$

The law, support gaps and sign split are those of Section 3; in particular D is the actual gap defect, not a moment surrogate. Write the central gap, once acquired, as $(-A_-, A_+)$, so that its endpoints are not confused with the sign indicator $B_0 = \mathbf{1}_{\{X>0\}}$.

Proposition 5.1 (Direct geometry)**.** *Let X be compactly supported, centered and of variance one. If $n \geq 700$ and $nD \leq B_*$, its closed support has a gap $(-A_-, A_+)$ across zero, with $A_-, A_+ > 0$. In the actual decomposition $X = \ell(B_0 + U - p)$, put*

$$a_0 = p - A_-/\ell, \quad a_1 = q - A_+/\ell, \quad d = a_0 + a_1,$$
$$\theta = \mathbb{E}\{X - \mathbb{E}(X \mid B_0)\}^2, \qquad \lambda = n\mathbb{E}U^2 = n\theta/\ell^2.$$

Then $a_0, a_1 \geq 0$, $\mathbb{E}(U \mid B_0) = 0$, $U_0 \leq a_0$, $U_1 \geq -a_1$, and

$$\frac{3}{8} < p < \frac{93}{200}, \qquad \theta < .0036, \qquad d < .004, \qquad \ell > 1.996, \tag{38}$$
$$\rho < \frac{9}{8}, \qquad .066148 < \beta < .2527, \qquad \lambda < \frac{7}{20}. \tag{39}$$

These conclusions require no contact or predecessor hypothesis. They express three consequences of a small defect: localization of the sign probability, a small inward displacement of the gap, and a small conditional noise. They will be proved from one common inequality, so the costs being combined all belong to the same actual marginal.

Proof. We have $D < D_0 := .047$, because $700D_0 > B_*$. If zero were in the closed support, Lemma 3.1 would give $D \geq (\sigma-1)\rho \geq \sigma - 1 > D_0$. Thus the central gap exists; centering supplies both signs. Conditional means imply $a_j \geq 0$ and $0 \leq d < 1$. Put $v = pq$ and $w = 1 - \theta = v\ell^2$, and define

$$Q(p) = (\sigma+1)p^2 + (\sigma-1)q^2 - 6pq = (6+2\sigma)(p-p_*)^2.$$

The common budget.

We first retain all the costs of the actual conditional noises. Let

$$\int f\, d\mu_\pm = \mathbb{E}\big[(X - \mathbb{E}(X \mid B_0))^2 \mathbf{1}_{\{\pm X>0\}} f(|X|)\big].$$

Their masses sum to θ. The outer-gap estimate gives the common budget

$$D \geq \ell w Q + E_0 + \int k_+\, d\mu_+ + \int k_-\, d\mu_-, \tag{40}$$

where

$$E_0 = 6\ell w\{d(1-d) + a_0 a_1\},$$
$$k_+(x) = (\sigma-1)(x + 2\ell q) + 6A_+, \qquad k_-(x) = (\sigma+1)(x + 2\ell p) + 6A_-.$$

Indeed

$$J_o^+ \le 2\int_{A_+}^{\infty}(t-A_+)\tau(t)\,dt = \mathbb{E}[X_+(X_+-A_+)^2],$$

and the negative outer gaps have the same expression in magnitudes. Add the central contribution $m(A_- + A_+)^2$ and use

$$\mathbb{E}X_+^3 = m(\ell q)^2 + \int (x+2\ell q)\,d\mu_+, \qquad R_+ = \mu_+(\mathbb{R}) + m(\ell q - A_+),$$

with the negative counterparts. Expansion yields (40). This also proves the budget for countably many gaps by monotone convergence; endpoint atoms do not alter the integrals.

Localize before sharpening the prices.

The quadratic Q measures departure from the distinguished two-point probability. The term E_0 prices the inward displacement, while the two integrals price the conditional variances. All are nonnegative. We first obtain a rough interval where every price is uniformly positive, and only then improve the individual bounds.

Hereafter use $3.162277660 < \sigma < 3.162277661$. The weaker gap estimate $J \le \rho + 2mA_-A_+$, together with $\rho \ge 1$, $\gamma \le \rho$ and $m \le 1/2$, gives

$$A_-A_+ \ge z := \frac{3.162277660 - 1 - D_0}{3} > 0, \qquad w \ge A_-A_+ \ge z.$$

The second inequality follows from $A_- \le \ell p$, $A_+ \le \ell q$. All costs in (40) are nonnegative, so

$$D \ge \frac{w^{3/2}Q}{\sqrt{v}} \ge 2z^{3/2}Q.$$

Let Q_- replace σ in its polynomial expression by the lower rational bound. The two exact tests

$$4z^3Q_-(.36)^2 > D_0^2, \qquad 4z^3Q_-(.48)^2 > D_0^2$$

exclude $p \le .36$ and $p \ge .48$, respectively: the squared distance to $p_* \in (.36, .48)$ is monotone outside that interval.

On $.36 < p < .48$, $\ell \ge 2\sqrt{z} > 1.67$, and the noise prices satisfy

$$k_+(x)/\ell \ge q\{(\sigma+5)z/w + 2(\sigma-1)\} > 5.2,$$
$$k_-(x)/\ell \ge p\{(\sigma+7)z/w + 2(\sigma+1)\} > 5.2.$$

For the first use $x \ge A_+$ and $A_+ \ge z/(\ell p)$; for the second use their negative counterparts. Since $w \le 1$, the constants follow by taking $q = .52$ and $p = .36$. Therefore $D \ge 1.67(5.2)\theta$, whence $\theta < .006$, $w > .994$ and $\ell > 1.99$. Also $A_- + A_+ \ge 2\sqrt{z}$ gives

$$d \le 1 - 2\sqrt{zv/w} < .2, \qquad v \ge .36(.64).$$

Evaluating the kernels at their lower support endpoints in (40) now yields

$$D \ge \ell\{wQ + 6wd(1-d) + \theta\min(C_+ - b_+d, C_- - b_-d)\}, \tag{41}$$

where

$$C_+ = 3(\sigma+1)q, \quad C_- = 3(\sigma+3)p, \quad b_+ = \sigma+5, \quad b_- = \sigma+7.$$

Each branch increases in d on the current domain, since $6(.994)(.6)-.006(3.162277661+7) > 0$; both noise prices are positive. The inward cost alone excludes $d \geq .004$, because

$$1.99(.994)6(.004)(.996) > .047.$$

After replacing d by zero, the derivative in θ of either branch $\sqrt{(1-\theta)/v}\{(1-\theta)Q+\theta C\}$ has the sign of $(2-3\theta)C-3(1-\theta)Q$. It is positive: $C > 6.49$, $Q < .05$ and $\theta < .006$. Hence $D \geq Q/\sqrt{v}$. This ratio decreases on $[.36, 3/8]$, and

$$Q_-(3/8)^2 > D_0^2(3/8)(5/8)$$

excludes that interval. On $[93/200, .48]$ instead use $Q \geq Q(93/200) > D_0/2$ and $\sqrt{v} \leq 1/2$. Thus $3/8 < p < 93/200$. Since

$$C \geq \min\{3(\sigma+1)(107/200),\ 3(\sigma+3)(3/8)\},$$

the bound $D \geq \ell\theta C$, with $\ell > 1.99$, improves $\theta < .0036$ and consequently $\ell > 1.996$.

The exact conditional cubic expansion gives

$$\rho = w^{3/2}\rho_B(p) + \int (x+2\ell q)\, d\mu_+ + \int (x+2\ell p)\, d\mu_-, \qquad \rho_B(p) = \frac{1-2v}{\sqrt{v}}.$$

Because the two integrands are priced by at least $\sigma - 1$ in (40),

$$\rho \leq \rho_B(p) + D/(\sigma-1) < 1.1 + .047/(54/25) < 9/8.$$

Here $v \geq 15/64$ implies $\rho_B < 1.1$ by squaring positive quantities. The signed-second-moment identity gives

$$1-2p-2q\theta \leq \beta \leq 1-2p+2p\theta;$$

its endpoints imply $.066148 < \beta < .2527$.

The noise budget in count coordinates.

The comparison needs $\lambda = n\theta/\ell^2$, so a bound on the one-variable variance θ alone is not the desired conclusion. The factor n is recovered by retaining the cubic cost rather than only its sign. If Z is centered, $Z \leq a$, $a > 0$, and $t = \mathbb{E}Z^2$, then

$$0 \leq \mathbb{E}[(a-Z)(Z+t/a)^2] = at - \mathbb{E}Z^3 - t^2/a.$$

Apply this to U_0 and $-U_1$. A zero endpoint forces the corresponding noise to vanish, so no division by zero is needed. For $\lambda_0 = nq\mathbb{E}U_0^2$, $\lambda_1 = np\mathbb{E}U_1^2$, the same gap-budget expansion, before bounding these two cubics, reads

$$\begin{aligned} nD/\ell^3 \geq\ & nvQ + 6nv\{d(1-d)+a_0a_1\} \\ & + \{3(\sigma+3)p-(\sigma+7)a_0\}\lambda_0 + \{3(\sigma+1)q-(\sigma+5)a_1\}\lambda_1 \\ & + \frac{(\sigma+1)\lambda_0^2}{nqa_0} + \frac{(\sigma-1)\lambda_1^2}{npa_1}. \end{aligned}$$

Terms with $a_j = 0$ are absent. Set $\bar{d} = .004$ and use $d(1-d)+a_0a_1 \geq (1-\bar{d})(a_0+a_1)$. Pair each displacement cost with its reciprocal cubic cost by AM–GM; for example

$$6nv(1-\bar{d})a_0 + \frac{(\sigma+1)\lambda_0^2}{nqa_0} \geq 2\lambda_0\sqrt{6(1-\bar{d})(\sigma+1)p}.$$

Thus

$$nD \geq \ell^3\{nvQ + \lambda_0 Y_0(p) + \lambda_1 Y_1(p)\}, \tag{42}$$

where

$$Y_0 = 3(\sigma+3)p - (\sigma+7)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma+1)p},$$
$$Y_1 = 3(\sigma+1)q - (\sigma+5)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma-1)q}.$$

Each increases in its respective argument p or q. At $p = 3/8$ and $q = 107/200$, lower bounds 3 and 2.6 for the roots give $Y_0 > 12.8$, $Y_1 > 11.8$. Since $\lambda_0 + \lambda_1 = \lambda$,

$$B_* \geq nD > 1.996^3(11.8)\lambda \quad (\lambda > 0), \qquad 1.996^3(11.8)(.35) > B_*.$$

This proves $\lambda < .35$; at zero noise it is immediate. □

Proposition 5.2 (Two contact rays)**.** *Under Proposition 5.1, suppose (P, H) is an attained fixed-target pair at C_{E} with nonnegative excess and true predecessor $C_{n-1} \leq C_{\mathrm{E}}$. Then*

$$-.6 < U_0 \leq a_0, \qquad -a_1 \leq U_1 < .6.$$

The support conclusion is uniform in the raw threshold H.

Proof. Use the contact upper expression (22). The all-threshold Gaussian heat estimate and its radial derivative bound from the proof of Theorem 4.1 give, for $t > \sqrt{3}$,

$$H_n(H, \pm t) \leq h(t) := S^3\left\{\frac{2}{5}\left(\frac{t^3-3t}{6} + \frac{|t^4-6t^2+3|}{72}\right) + \frac{t^5}{4100} + \frac{3t}{820} + \frac{17(3t^2-2)}{78720}\right\},$$
$$|\partial_t H_n(H, \pm t)| < g(t) := \frac{2}{5}(t^2/2 + 1/2 + \eta), \qquad S = 81/80, \quad \eta = 8/875.$$

The first bound follows directly from (23): $\|\phi'''\|_\infty < 2\phi(0)$, $n - 1 \geq 41$, and $\sqrt{n-1} > 6$. No derivative of the absolute-value majorant is taken.

At a support point of magnitude t, contact requires a nonnegative upper expression $H_n(H, \pm t) + C_{\mathrm{E}} Q_b(t)$, where

$$Q_b(t) = -t^3 + \tfrac{3}{2}rt^2 + 3bt + \eta r, \quad r = 9/8, \quad b = \begin{cases} -.066148 & \text{negative ray,} \\ .2527 & \text{positive ray.} \end{cases}$$

These replacements follow with their signs from (39); the discarded excess term is nonpositive. Take $c_0 = 819/2000 < C_{\mathrm{E}}$ and the two fixed boundaries

$$t_- = 1.996(3/8 + .6) = 1.9461, \qquad t_+ = 1.996(107/200 + .6) = 2.26546.$$

For the respective b, direct rational substitution at each boundary gives

$$t^2 > 3, \qquad h(t) + c_0 Q_b(t) < 0, \qquad Q_b, Q_b', Q_b'' < 0, \qquad L, L', L'' < 0, \quad L = g + c_0 Q_b'.$$

The two boundary upper bounds are less than $-.40$ and $-.13$, respectively. These are just polynomial comparisons at two rational points. Since $Q_b''' = -6$ and $L'' = 2/5 - 6c_0 < 0$, the required signs persist along both outward rays. The derivative of the actual upper expression is below L, as $Q_b' < 0$ and $C_{\mathrm{E}} > c_0$. It therefore stays strictly negative beyond either boundary. Contact excludes every support point there, including boundary atoms.

On the negative class, $X > -t_-$ and $\ell > 1.996$ give $U_0 = X/\ell + p > -(3/8 + .6) + p > -.6$. On the positive class, $X < t_+$ gives $U_1 = X/\ell + p - 1 < 107/200 + .6 + p - 1 < .6$. The inward endpoints and conditional centering are the actual ones from Proposition 5.1. □

Return to the original marginal.

The two ray bounds concern the conditional supports of the original law. They do not introduce a new optimizing distribution. The pointwise normalization is worth recording once:

$$1 = v\ell^2 + \theta, \qquad s = \mathbb{E}U^2 = \theta/\ell^2, \qquad v + s = \ell^{-2}, \qquad X = \frac{B_0 + U - p}{\sqrt{v+s}}. \tag{43}$$

Thus the law to be compared in the next section is exactly the original law, and its threshold remains the raw H. Reflection, when required, selects the oriented positive witness before acquisition; neither its signed defect budget nor its probability interval is transported from the opposite orientation.

6 The direct comparison with a binomial law

The acquired strip admits a single comparison on its whole probability interval. Very small noise is controlled by the loss at one count and fourth moments. For the rest, five adjacent counts give a uniform crossing loss. Both arguments concern the actual conditional laws of one marginal.

Theorem 6.1 (The common-strip comparison)**.** *Let $n \geq 700$ be an integer, $B \sim \mathrm{Bernoulli}(p)$, and $\mathbb{E}(U \mid B) = 0$, where $p \in [3/8, 93/200]$. Write U_j for the conditional noise given $B = j$, and suppose*

$$-3/5 \leq U_0 \leq a_0, \qquad -a_1 \leq U_1 \leq 3/5, \qquad a_0, a_1 \geq 0, \quad a_0 + a_1 \leq 1/250.$$

Put $q = 1 - p$, $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (B + U - p)/\sqrt{v+s}$ and $\rho = \mathbb{E}|X|^3$. For independent copies and every raw threshold $|H| \leq 46/25$,

$$\sqrt{n}\left\{\mathbb{P}\left(\sum_{i=1}^{n} X_i \leq H\right) - \Phi(H/\sqrt{n})\right\} - C_{\mathrm{E}}\rho < -\lambda/100 \qquad (0 < \lambda \leq 7/20). \tag{44}$$

At $\lambda = 0$ the same excess is strictly negative.

The proof below uses only the displayed hypotheses. Contact and the sharp predecessor are needed earlier to acquire this strip, not for its local comparison. The small- and larger-noise intervals meet at $\lambda = 1/100$. The comparison is relative to the noise: its gain is at least $\lambda/100$ for positive noise. This is why the estimates must retain small terms uniformly as λ tends to zero. The exact zero-noise case will be handled separately at the end.

6.1 Three elementary ingredients

Write $\pi_j = \binom{n}{j} p^j q^{n-j}$, $F_B(j) = \sum_{i \leq j} \pi_i$, and $\rho_B = (1 - 2v)/\sqrt{v}$. We use Schulz's strict finite-binomial theorem [4, Theorem 1] in the form

$$\sqrt{n}\{F_B(j) - \Phi((j - np)/\sqrt{nv})\} < C_{\mathrm{E}}\rho_B \quad (0 < p < 1, n < \infty). \tag{45}$$

Its strictness concerns every finite binomial law; it is not inferred from the limiting lower approximation for the universal constant.

The factorial estimates used throughout are finite Stirling bounds:

$$\log(j!) = (j+\tfrac{1}{2})\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j).$$

For completeness, with $x = 1/(2j+1)$, $r_j - r_{j+1} = \operatorname{arctanh}(x)/x - 1 = \sum_{l\ge1} x^{2l}/(2l+1)$ lies strictly between 0 and $1/(12j(j+1))$. Wallis's product fixes the limiting constant, so $r_j \to 0$; summing these differences proves the bounds. One useful consequence is

$$\max_j \mathbb{P}\{\mathrm{Bin}(M,p) = j\} < \frac{41}{100\sqrt{Mv}} \quad (M \ge 43, p \in [1/3, 2/3]). \tag{46}$$

Indeed a mode has $z = j/M = p + \delta/M$, $|\delta| \le 2/3$. Since $v \ge 2/9$ and $|1-2p| \le 1/3$, $z(1-z) \ge v(1 - 1/M - 2/M^2) > 39v/40$. Stirling, the nonnegative relative entropy, and $e^{1/516} < 516/515$ bound the mass by $(2/5)\sqrt{40/39}(516/515)/\sqrt{Mv}$, which is smaller than the right side of (46) by a rational square comparison.

Finally the cubic correction has the favorable sign on the full strip. Let $s_j = \mathbb{E}U_j^2$, so $s = qs_0 + ps_1$, and set $r = 3/8$, $d = 1/250$, $v_* = 15/64$. The conditional supports have fixed signs relative to p, since $d < r$. Expansion of the raw third absolute moment gives

$$\mathbb{E}|B + U - p|^3 = v(1-2v) + 3v(s_0+s_1) - q\mathbb{E}U_0^3 + p\mathbb{E}U_1^3.$$

The inward bounds imply $\mathbb{E}U_0^3 \le a_0 s_0$, $\mathbb{E}U_1^3 \ge -a_1 s_1$. Since $p, q \ge r$ and $a_j \le d$, the variance terms satisfy

$$3v(s_0+s_1) - qa_0 s_0 - pa_1 s_1 \ge (3r - d)s.$$

Standardizing the noisy law also changes the factor multiplying ρ_B. Its exact contribution is

$$\rho_B\{(v+s)^{3/2} - v^{3/2}\} = v(1-2v)\{(1+s/v)^{3/2} - 1\}.$$

Thus a lower bound for the raw third moment and an upper bound for this normalization cost have to be combined before discarding the correction. Together with $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$, this yields

$$\begin{aligned}(v+s)^{3/2}(\rho - \rho_B) &\ge s\left\{3r - d - \frac{3}{2}(1-2v) - \frac{3(1-2v)s}{8v}\right\} \\ &\ge s\left\{\frac{2593}{8000} - \frac{17}{20}s\right\} > 0, \qquad 0 < s \le \frac{1}{2000}.\end{aligned} \tag{47}$$

Here $s \le (7/20)/700 = 1/2000$, and the displayed coefficient is $3r - d - \frac{3}{2}(1-2v_*)$. Thus every nonnegative coefficient of $\rho_B - \rho$ can be discarded in an upper comparison.

6.2 Small noise: one count and the whole incoming tail

Assume $0 < \lambda \le L = 1/100$. Choose an integer k with

$$t = np + H\sqrt{v+s} = k + u, \qquad -3/10 \le u \le 7/10, \qquad \delta = k - np.$$

The phase interval has length one. It need not use the same anchor as the larger-noise proof. Let W_j have the law of the sum of j independent U_1 variables and $n-j$ independent U_0 variables. For $C \ge 0$, subtract the signed binomial excess at k from the actual excess:

$$\begin{aligned}\Delta_C = \sqrt{n}\left\{\mathbb{P}\left(\sum X_i \le H\right) - \Phi(H/\sqrt{n})\right\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B.\end{aligned}$$

Exact conditioning gives

$$\mathbb{P}\left(\sum X_i \le H\right) - F_B(k) = -\sum_{i=0}^{k} \pi_{k-i}\mathbb{P}(W_{k-i} > u+i) + \sum_{i=1}^{n-k} \pi_{k+i}\mathbb{P}(W_{k+i} \le u-i).$$

The first sum records mass which had been below the threshold at a binomial count and is pushed out by noise. The second records mass pushed in from a higher count. Outgoing events are strict and incoming events closed; this convention includes all threshold atoms. We will keep the loss at count k, and bound every incoming count. Define

$$f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}, \qquad O = \mathbb{P}(W_k > u) + f(u),$$

$$T_{\text{in}} = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u-i), \qquad \mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

Keeping the own-count loss and discarding the other nonpositive outgoing terms gives

$$\Delta_C \le -m_k(O - RT_{\text{in}}) + \mathcal{R}, \qquad m_k = \sqrt{n}\pi_k, \tag{48}$$

whenever $\pi_{k+i}/\pi_k \le R$ for all legal incoming counts.

Uniform normalization.

Set $D_* = 5/3$, $b_* = 1/20$, $T_* = 101/10000$ and $c_* = 2/525$. Because $v \ge v_*$ and $v + s < 1/4$, $|t - np| < 23/25$. Consequently $|\delta| < D_*$ globally and $|\delta| < 1$ when $|u| \le b_*$. Write $\mathcal{K}(z,p) = z\log(z/p) + (1-z)\log((1-z)/q)$ for binomial relative entropy; this is distinct from the physical gap defect D. The segment between p and k/n lies inside $[.37, .47]$; in particular $1 \le k \le n-1$. On this segment

$$z(1-z) > 23/100, \quad |\partial_z^3 \mathcal{K}(z,p)| < 5, \quad |(\log(z(1-z)))'| < 6/5, \quad (12z(1-z))^{-1} < 3/8.$$

Stirling and entropy Taylor expansion therefore give

$$m_k = \frac{\phi(0)}{\sqrt{v}} e^{-\delta^2/(2nv)+e}, \qquad n|e| < Z(D_0) := 3D_0/5 + 3/8 + D_0^3/840 \tag{49}$$

for $|\delta| \le D_0 \le D_*$. For clarity, if $z_0 = k/n$, the remainder in this display is exactly

$$e = -\frac{1}{2}\log\frac{z_0(1-z_0)}{v} - n\left\{\mathcal{K}(z_0,p) - \frac{(z_0-p)^2}{2v}\right\} + r_n - r_k - r_{n-k}.$$

The three terms in $Z(D_0)$ respectively bound the logarithmic prefactor, factorial remainder and entropy remainder. The segment bounds above therefore control this error throughout the allowed phase interval. Also

$$\log f'(z) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+z)^2}{2n(v+s)} - e,$$

so, on an integration interval $|z| \le b_0$,

$$n|\log f'(z)| \le \frac{11}{5}\lambda + \frac{2D_0 b_0 + b_0^2}{2v_*} + Z(D_0).$$

The coefficient $11/5$ exceeds $1/(2v_*) + D_*^2/(1400v_*^2)$. Using $e^{-x} \geq 1-x$, $e^x \leq (1-x)^{-1}$ and $\phi(0) > 199/500$ now gives

	$\lvert u\rvert \leq 1/20$	$-3/10 \leq u \leq 7/10$
m_k lower bound	$3/4$	$3/4$
f' lower bound	$99/100$	$49/50$
f' upper bound	$101/100$	$51/50$
R upper bound	$51/50$	$6/5$

(50)

The derivative estimates hold on the whole segment from 0 to u. For the global ratio, (46) gives $\sqrt{n}\max_j \pi_j < 17/20$, hence $R < (17/20)/(3/4) < 6/5$. In the inner interval the first ratio is

$$\frac{\pi_{k+1}}{\pi_k} = 1 + \frac{p-\delta-1}{nv + q(\delta+1)} < 1 + \frac{93/200}{700v_*} < 51/50;$$

every later ratio is at most one, since $p-\delta < 2$. By (47) the moment correction is nonpositive. Differentiating the normal CDF and using $\phi(0)/(2v_*^{3/2}) < 2$ therefore gives

$$\mathcal{R} < 2\lambda/n \quad (|u| \leq b_*), \qquad \mathcal{R} < 4\lambda/n \quad \text{on the full phase interval.} \tag{51}$$

Indeed the positive displacement is bounded by 1 or $D_* < 2$.

No truncation of the incoming tail.

Since $|s_1 - s_0| \leq (8/3)s$, the actual own-count variance $t_0 = \mathbb{E}W_k^2$ satisfies $.99\lambda < t_0 < 1.01\lambda \leq T_*$. For each incoming count,

$$t_i = \mathbb{E}W_{k+i}^2 \leq \lambda\{1 + c_*(D_0 + i)\}, \qquad \mathbb{E}W_{k+i}^4 \leq A^2 t_i + 3t_i^2, \quad A = 3/5,$$

where $D_0 = 1$ in the inner interval and $D_0 = D_*$ otherwise. The moment bound follows by expanding the fourth power of a sum of independent centered variables: the fourth individual moments cost at most $A^2 t_i$, and the pairings cost at most $3t_i^2$. The variance changes with the incoming count, which is why its bound retains the term linear in i. For $a = 1 + c_* D_0$ put

$$K(a) = A^2\left(\frac{13a}{12} + \frac{5c_*}{4}\right) + 3L\left(\frac{13a^2}{12} + \frac{5ac_*}{2} + \frac{5c_*^2}{3}\right).$$

To see the origin of $K(a)$, for $0 \leq u < 1$ use $i - u \geq (1-u)i$, $\lambda \leq L$, and fourth-moment Markov. The resulting majorant, divided by $\lambda/(1-u)^4$, is

$$A^2\left\{a\sum_{i\geq 1} i^{-4} + c_*\sum_{i\geq 1} i^{-3}\right\} + 3L\left\{a^2\sum_{i\geq 1} i^{-4} + 2ac_*\sum_{i\geq 1} i^{-3} + c_*^2\sum_{i\geq 1} i^{-2}\right\}.$$

The stated series bounds give $K(a)$. In particular, the estimate covers the entire incoming tail even though only finitely many counts are legal. Using the corresponding denominator bound for each phase region gives

$$\begin{aligned} T_{\text{in}} &< \frac{K(1+c_*)}{(1-b_*)^4}\lambda < .53\lambda \quad && (|u| \leq b_*), \\ T_{\text{in}} &< \frac{.43\lambda}{(1-u)^4} && (0 \leq u \leq .7), \\ T_{\text{in}} &< .43\lambda && (u \leq 0). \end{aligned} \tag{52}$$

Here $K(1+c_*D_*) < .43$, and $\sum i^{-4} < 13/12$, $\sum i^{-3} < 5/4$, $\sum i^{-2} < 5/3$. These series bounds follow by retaining respectively five, three and five terms and estimating the remaining sum by an integral. For $u \geq 0$ use $i - u \geq (1-u)i$; for $u \leq 0$ use $i - u \geq i$.

The own-count loss at the central phases.

Near $u = 0$, the normal increment alone gives no uniform positive reserve. We therefore bound the strict outgoing probability directly by a polynomial minorant, whose expectation can be expressed through the second, third and fourth moments. The value of that polynomial at zero must respect the strict-event convention.

With $\alpha = 13/25$ and $M = 5\alpha^4$, the quartic

$$h_\alpha(x) = \frac{-x^4 + 3\alpha^2 x^2 + 2\alpha^3 x}{M} \le \mathbf{1}_{(0,\infty)}(x)$$

is a strict-event minorant: its numerator factors as $-x(x+\alpha)^2(x-2\alpha)$, and its positive maximum divided by M is $(9+6\sqrt{3})/20 < 1$. Let $r = zt_0$ and set

$$F_{\varepsilon,b}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{M} + \left(\frac{2\varepsilon}{5\alpha} + b - \frac{4At}{M}\right) z + \frac{3\alpha^2 t - 6t^2}{M} z^2 - \frac{t^3 z^4}{M}.$$

Expanding $h_\alpha(W_k + r)$ for $u = -r$, or $h_\alpha(W_k - r)$ for $u = r$, and using $|\mathbb{E}W_k^3| \le At_0$, $\mathbb{E}W_k^4 \le A^2 t_0 + 3t_0^2$, yields

$$O/t_0 \ge F_{1,-101/100}(z,t_0) \quad (u=-r), \qquad O/t_0 \ge F_{-1,99/100}(z,t_0) \quad (u=r).$$

Both functions exceed $3/5$ on $0 \le z \le 3/2$, $0 \le t \le T_*$. To check the whole rectangle, use concavity in t and check only $t = 0, T_*$. At those endpoints replace z^4 by $(9/4)z^2$ in its negative term. The negative-phase quadratic decreases on $[0, 3/2]$, and the positive-phase quadratic increases; their minima are respectively at $z = 3/2$ and $z = 0$. All four rational endpoint values exceed $3/5$.

For negative phases $3t_0/2 \le r \le b_*$, Cantelli gives

$$O \ge \frac{r^2}{t_0 + r^2} - \frac{101}{100} r > \frac{3}{5} t_0.$$

Its derivative at $r = 3t_0/2$ is positive and first increases, then decreases. The minimum is therefore at an endpoint. At the first use $t_0 \le T_*$ after division by t_0; at the other use $b_*^2/(T_* + b_*^2) - (101/100)b_* > (3/5)T_*$. For positive phases beyond this switch, $O \ge (99/100)r \ge (297/200)t_0 > (3/5)t_0$. The switch lies below b_* because $3T_*/2 < b_*$. Thus

$$O > \frac{3}{5} t_0 > .59\lambda \qquad (|u| \le b_*). \tag{53}$$

The value $u = 0$ is included, with $h_\alpha(0) = 0$.

The outer phases and the final margin.

For $u = -r$, $b_* \le r \le 3/10$, Cantelli gives

$$O \ge \frac{r^2}{T_* + r^2} - \frac{51}{50} r > 7/50 \ge 14\lambda.$$

The derivative is positive at b_* and has at most one downward zero; both endpoint values exceed $7/50$. For $b_* \le u \le 7/10$,

$$O - RT_{\text{in}} \ge \frac{49}{50} u - \frac{(6/5)(43/100)L}{(1-u)^4} > 1/25 \ge 4\lambda.$$

This function is concave and both endpoints exceed 1/25. In the inner interval, (52) and (53) give $O - RT_{\mathrm{in}} > (.59 - (51/50).53)\lambda$. Substitute these three margins and (51) into (48), using $m_k > 3/4$. The smallest final margin is the inner one, and

$$\frac{3}{4}\left(\frac{59}{100} - \frac{51}{50}\frac{53}{100}\right) - \frac{2}{700} > \frac{1}{100}.$$

Hence $\Delta_C < -\lambda/100$. Taking $C = C_{\mathrm{E}}$ and applying (45) proves (44) for $0 < \lambda \le 1/100$.

6.3 Larger noise: five counts and four endpoint checks

Now assume $1/100 \le \lambda \le 7/20$ and choose $k = \lfloor t \rfloor$, $u = t - k \in [0, 1)$, $w = t - np$. The proof has three parts. First, a summation identity separates a nonnegative crossing loss from a curvature error. Next, five adjacent counts give a uniform lower bound for the loss. Finally, the complete curvature tail and the normal correction are paid from that reserve. The following constants will be used only in this subsection:

$$\begin{gathered} N = 700, \quad p_- = 3/8, \quad p_+ = 93/200, \\ r = 3/8, \quad q_* = 107/200, \\ v_- = 15/64, \quad v_+ = 9951/40000, \quad A = 3/5, \quad d = 1/250, \\ \eta = 2/5, \quad \beta_0 = 41/100, \quad \pi_+ = 22/7, \quad \phi_- = 199/500, \quad \phi_+ = 2/5, \\ \kappa = 3, \quad \xi = 93/100, \quad c_- = 6, \quad \zeta = 23/100, \quad m_* = 3/4, \quad V_* = 113/1000, \\ \bar{c} = 79/200, \quad j_0 = 1/5, \quad j_1 = 5/2. \end{gathered}$$

In particular $|w| < \xi < 1$, since $(46/25)^2(v_+ + (7/20)/N) < \xi^2$. Also $N(r - d) > \xi$, so $na_0 \le t < n(1 - a_1)$.

An exact weighted crossing identity.

On an auxiliary space take independent copies of the conditional laws and write

$$T_j = j + \sum_{l \le j} U_{1l} + \sum_{l > j} U_{0l}, \qquad L_j(t) = \mathbb{E}(t - T_j)_+, \qquad V_i = \sum_{l < i} U_{1l} + \sum_{l > i} U_{0l}.$$

The mesh is increasing because $T_j - T_{j-1} \ge 1 - a_0 - a_1 > 0$. Each T_j has the conditional convolution given j successes. Set

$$h(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \le x), \qquad g_i = \mathbb{E}h(t - i + 1 - V_i).$$

Independence of the omitted coordinate gives $g_i = L_{i-1}(t) - L_i(t) - \mathbb{P}(T_i \le t)$. The support assumptions imply $T_0 \le na_0 \le t$ and $T_n \ge n(1 - a_1) > t$. Since $\mathbb{E}T_0 = 0$, these endpoint facts give $L_0 = t$, $L_n = 0$ and $\mathbb{P}(T_0 \le t) = 1$. Thus no boundary term is lost when the identity is summed. Multiplication by π_i and finite summation by parts yield

$$\begin{aligned} \mathbb{P}\left(\sum X_i \le H\right) &= (1 - u)F_B(k) + uF_B(k + 1) \\ &\quad + \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j)B_j - \sum_{i=1}^{n} \pi_i g_i, \quad B_j = L_j(t) - (t - j)_+ \ge 0. \end{aligned} \tag{54}$$

Indeed replacing L_j by $(t - j)_+$ in $\pi_0 + \pi_1 t + \sum(\pi_{j+1} - \pi_j)L_j$ telescopes to the stated interpolation. All CDF events are closed, including atoms.

Five kernels with four centered replacements.

Let $f(x) = \sin^2(\pi x)$ on $[0,1]$ and zero outside. It is continuously differentiable with derivative Lipschitz constant $2\pi^2$. We claim

$$h(x) \geq \eta f(x) \geq 0. \tag{55}$$

For $x < 1 - a_1$, $h(x) = \mathbb{E}(x - U_0)_+ \geq 0$, and Jensen gives $h(x) \geq x$ when $x \geq 0$. On $[0,1]$, $f(x) \leq (5/2)x$: use $\pi^2 < 10$ up to $1/4$, the concave tangent $1/2 + \pi(x - 1/4)$ on $[1/4, 2/5]$, and $f \leq 1$ thereafter. For $x \geq 1 - a_1 > a_0$, put $D_1 = 1 + U_1$ and $L_1 = \mathbb{E}(D_1 - x)_+$. Put–call parity gives $h(x) = \mathbb{P}(D_1 > x) - L_1 \geq 0$, since $D_1 \leq 1 + A$ and $x > A$. When $1 - a_1 \leq x \leq 1$, $L_1 \geq 1 - x$ and

$$h(x) \geq \frac{x - A}{1 + A - x} L_1 \geq \frac{1 - A - d}{A + d}(1 - x) \geq \eta\pi^2(1 - x)^2 \geq \eta f(x).$$

The penultimate inequality uses $\eta\pi_+^2 d \leq (1 - A - d)/(A + d)$. At $x = 0, 1$ both comparisons are weak, as required for atoms.

Freeze $V = V_{k+1}$ and let $V^{(j)}$ have the law of V_{k+1+j}. For $j > 0$ it replaces j copies of U_0 by U_1. With $f_j(x) = f(u - j - x)$, regroup the two positive changes as

$$\begin{aligned}
&\mathbb{E}f_1(V^{(1)}) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_1(V) - \mathbb{E}f_2(V) \\
&\quad = \mathbb{E}(f_1 + f_2)(V^{(1)}) - \mathbb{E}(f_1 + f_2)(V) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_2(V^{(1)}).
\end{aligned}$$

There is an identical grouping for the negative changes. Adjacent bumps have disjoint open supports and zero derivatives at the joins, so the derivative Lipschitz constant remains $2\pi^2$. For a centered Z, $|\mathbb{E}F(a + Z) - F(a)| \leq \frac{1}{2}\operatorname{Lip}(F')\mathbb{E}Z^2$. Condition on all unchanged summands. Replacing one centered U_0 by one centered U_1 can be bounded by removing the first variable to its mean and then adding the second. The two Taylor remainders together cost at most $\pi^2(s_0 + s_1)$. The displayed regrouping and its negative counterpart need four such replacements, for a total $4\pi^2(s_0 + s_1)$. Hence

$$\sum_{i=k-1}^{k+3} g_i \geq \eta \left\{ \sum_{j=-2}^{2} \mathbb{E}f(u - j - V) - 4\pi^2(s_0 + s_1) \right\}.$$

These five indices are interior, because $|i - np| < \xi + 3 < 4$. No independence between different count sums is asserted.

The infinite translates of f sum to $\sin^2(\pi(u - V))$. An omitted nonzero translate requires $|V| > 2$, including endpoint collisions because the bump vanishes there. Thus the omitted part is at most $\mathbb{E}V^4/16$. With $\delta = k - np \in (-1 - \xi, \xi)$, the frozen variance is

$$\tau = (nq - \delta - 1)s_0 + (np + \delta)s_1, \qquad (1 - 6/n)\lambda \leq \tau \leq (1 + 3/n)\lambda.$$

The lower estimate uses $c_- r \geq 1 + \xi$; the upper uses $r \geq 1/3$. Also $s_0 + s_1 \leq \lambda/(nr)$ and $\mathbb{E}V^4 \leq A^2\tau + 3\tau^2$. For either conditional noise Z, an independent copy satisfies $|Z - Z'| \leq A + d < 2/3$. Concavity of sine and $\sin(2\pi/3) = \sqrt{3}/2 > 17/20$ give $\sin(\pi x) \geq (51/40)x$ on $[0, A + d]$; hence $1 - \cos(2\pi x) \geq 3x^2$ on $|x| \leq A + d$. Therefore

$$1 - |\mathbb{E}e^{2\pi i Z}|^2 = \mathbb{E}\{1 - \cos(2\pi(Z - Z'))\} \geq 2\kappa \operatorname{Var} Z,$$

and $|\mathbb{E}e^{2\pi i Z}| \leq e^{-\kappa \operatorname{Var} Z}$. Multiplying over the independent frozen summands shows $\mathbb{E}\sin^2(\pi(u - V)) \geq (1 - e^{-\kappa\tau})/2$. Altogether,

$$\sum_{i=k-1}^{k+3} g_i \geq \eta \left\{ \frac{1 - e^{-\kappa(1-6/n)\lambda}}{2} - \frac{A^2(1 + 3/n)\lambda + 3(1 + 3/n)^2\lambda^2}{16} - \frac{4\pi^2\lambda}{nr} \right\}. \tag{56}$$

The weights and the complete curvature tail.

For $k-1 \le i \le k+3$, the segment between i/n and p has variance product greater than ζ; indeed $\zeta < \min_{z=p_- -4/N,\, p_+ +4/N} z(1-z)$. Relative entropy is bounded by $(i-np)^2/(2n\zeta) < 8/(n\zeta)$, and the factorial remainder is greater than $-1/(12n\zeta)$. Stirling therefore gives

$$\sqrt{n}\pi_i > 2\phi_- \left(1 - \frac{8+1/12}{N\zeta}\right) > m_*.$$

Every other g_i is nonnegative. The reserve in (56) increases with n. Replace n by N first, then use $1-e^{-x} \ge x - x^2/2 + x^3/6 - x^4/24$ for $x \ge 0$. Writing $\alpha = 1-6/N$, $b = 1+3/N$ and $z = \kappa\alpha$, we obtain

$$\sqrt{n}\sum_i \pi_i g_i \ge W(\lambda) := \eta m_* \left\{ \frac{z\lambda - z^2\lambda^2/2 + z^3\lambda^3/6 - z^4\lambda^4/24}{2} - \frac{A^2 b\lambda + 3b^2\lambda^2}{16} - \frac{4\pi_+^2\lambda}{Nr} \right\}. \tag{57}$$

Put $c = (np - q - t)_+$. For $a = t - j$ and $W_j = T_j - j$, centering gives

$$B_j = (\mathbb{E}|a - W_j| - |a|)/2 \ge 0, \qquad B_j \le \sqrt{T}/2, \qquad a_+ B_j \le T/4,$$

whenever $T \ge \mathbb{E}W_j^2$. The factorization $27x^4 - 256x + 256 = (3x-4)^2(3x^2+8x+16)$ gives $B_j \le 27M/(256|a|^3)$ for $a \ne 0$, if $M \ge \mathbb{E}W_j^4$. These bounds have complementary uses. The first follows from the Lipschitz property of absolute value and Cauchy; the second uses $4a(x-a)_+ \le x^2$ for $a > 0$. The fourth-moment bound prices distant counts at order $|a|^{-3}$. We retain all three because the nearest mesh points and the full tail require different estimates. The exact adjacent-mass formula is

$$\pi_{j+1} - \pi_j = \frac{np - q - j}{(n+1)v}\, \mathbb{P}\{\mathrm{Bin}(n+1, p) = j+1\}.$$

Discard its nonpositive terms and bound the positive numerator by $(a+c)_+$. Since $c \le \bar{c} < 1$, at most one mesh point lies in each of $(-c, 0)$ and $[0, 1)$; together they cost $c\sqrt{T} + T/4$. The rest is bounded by the complete series $\sum i^{-2} < 5/3$, $\sum i^{-3} < 5/4$. Using (46) and $\sqrt{n}/(n+1)^{3/2} \le 1/n$ yields

$$\sqrt{n}\sum_j (\pi_{j+1} - \pi_j) B_j \le \frac{\beta_0}{nv^{3/2}} \left\{ \frac{T}{4} + c\sqrt{T} + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right) \right\}. \tag{58}$$

A positive adjacent mass requires $j < np$, where its actual variance is at most $\lambda/q \le T = \lambda/q_*$ and $\mathbb{E}W_j^4 \le M = A^2T + 3T^2$. Moreover $J(\lambda) = j_1\lambda + j_0 \ge \sqrt{T}$, since $4j_0j_1 \ge 1/q_*$.

Both signs of the normal correction.

There are two normal errors to control: interpolation between neighboring counts and the variance change from v to $v+s$. Their signs depend on the raw threshold. Keeping them separate avoids paying for a favorable change. Apply (45) at $k, k+1$ and interpolate. For $G(x) = \Phi((x-np)/\sqrt{nv})$, $G''(x) \le \phi(0)(np-x)_+/(nv)^{3/2}$. The interpolation remainder is

$$\sqrt{n}\{(1-u)G(k) + uG(k+1) - G(t)\} \le \frac{\phi(0)}{2nv^{3/2}} u(1-u)(u-w)_+.$$

If $H \le 0$, changing normal variance from v to $v+s$ is favorable. Using $u(1-u) \le 1/4$ and $u^2(1-u) \le 4/27$, the total normal cost is at most $\phi_+(\xi/8 + 2/27)/(nv^{3/2})$, and $c =$

$(-q-w)_+ \le \bar{c}$. If $H \ge 0$, $c = 0$, interpolation costs at most $\phi_+(2/27)/(nv^{3/2})$, and the variance change costs at most $\phi(0)ws/(2v^{3/2})$ by $1-(1+x)^{-1/2} \le x/2$. The cubic correction is favorable by (47).

Define the nonnegative quadratics

$$S_c = \frac{T}{4} + cJ(\lambda) + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right), \qquad E_- = \frac{\beta_0 S_{\bar{c}} + \phi_+(\xi/8 + 2/27)}{V_*},$$

$$E_+ = \frac{\beta_0 S_0 + \phi_+(2/27 + \xi\lambda/2)}{V_*}.$$

Because $V_*^2 < v_-^3$, combining (54)–(58) with these normal bounds makes the full excess strictly less than $-W(\lambda) + E_\pm(\lambda)/n$. It remains to compare these explicit polynomials.

Four rational endpoints finish the interval.

Set $P_\pm(\lambda) = W(\lambda) - E_\pm(\lambda)/N - \lambda/100$. Each $E_\pm$ is a convex quadratic, whereas direct differentiation gives

$$W''(\lambda) = -\eta m_* \left\{ \frac{z^2}{4}\left[(z\lambda - 1)^2 + 1\right] + \frac{3b^2}{8} \right\} < 0.$$

Thus $P_\pm$ is strictly concave. Substitution of the rational constants above gives the following exact rational lower bounds:

	$\lambda = 1/100$	$\lambda = 7/20$
P_-	.002314	.061304
P_+	.003341	.063680

All four are positive. More explicitly, put $a = 1/100$, $b = 7/20$ and $\omega = (\lambda - a)/(b - a)$. Concavity gives

$$P_\pm(\lambda) \ge (1-\omega)P_\pm(a) + \omega P_\pm(b) > 0 \qquad (a \le \lambda \le b).$$

This is the continuous-interval argument behind the four endpoint tests. Since $n \ge N$ and $E_\pm \ge 0$, this proves (44) for $1/100 \le \lambda \le 7/20$. The scalar checks used here involve only rational inequalities and $199/500 < \phi(0) < 2/5$, $25/8 < \pi < 22/7$; no parameter grid or tail truncation is involved.

The two positive-noise proofs have identical support and probability hypotheses, include their shared endpoint, and cover every phase. At $\lambda = 0$, conditional centering and $\mathbb{E}U^2 = 0$ give $U = 0$ almost surely, so (45) applies directly. This completes the proof of Theorem 6.1.

7 Composition and the finite boundary

We now separate the local exclusion from the induction that could make it global. The local theorem in Section 6 applies to every marginal satisfying its explicit strip assumptions. Contact, the predecessor bound and the entry predicates are used to show that an attained nonnegative-excess witness is one of those marginals.

Theorem 7.1 (Acquired-witness exclusion)**.** *Let $n \ge 700$ and let (P, H) attain the fixed-target maximum at C_{E} with $a(P, H) \ge 0$. Assume $C_{n-1} \le C_{\mathrm{E}}$, the structural inequality* (21)*, and complete entry Conditions 4.3 and 4.5. Then such a pair cannot exist.*

Proof. Theorem 4.6 gives

$$nD \le 32.20665378, \qquad |H| \le 1.83605680$$

for the selected oriented law. Proposition 5.1 then gives its actual central gap, $3/8 < p < 93/200$, inward sum $a_0 + a_1 < 1/250$ and $\lambda < 7/20$. The two contact rays of Proposition 5.2 give conditional amplitude $3/5$. This is where the sharp predecessor enters the support acquisition.

Equation (43) identifies the standardized Bernoulli-plus-noise marginal with P pointwise. Its threshold is the same raw H, and $1.83605680 < 46/25$. Thus all hypotheses of Theorem 6.1 hold simultaneously for this marginal. At positive noise the theorem gives

$$a(P, H) < -\lambda/100 < 0.$$

At zero noise the marginal is Bernoulli and the strict finite-binomial theorem gives $a(P, H) < 0$ directly. Both cases contradict the assumed nonnegative excess. □

Theorem 7.2 (Conditional global conclusion)**.** *Assume $C_j \le C_{\mathrm{E}}$ for $1 \le j \le 699$, the stated external structural and binomial theorems, and the complete entry predicates at every least failure above this base. Then*

$$C_n < C_{\mathrm{E}} \quad (n \ge 700), \qquad C_n \le C_{\mathrm{E}} \quad (n \ge 1), \qquad \sup_{n \ge 1} C_n = C_{\mathrm{E}}.$$

Proof. Suppose the strict inequality fails at some $n \ge 700$, and choose the least such index n_0. The base controls every index below 700; minimality controls each index from 700 to $n_0 - 1$. In particular $C_{n_0-1} \le C_{\mathrm{E}}$, including when $n_0 = 700$.

Failure includes the equality case $C_{n_0} = C_{\mathrm{E}}$. The attainment and reflection argument in Section 2.2 therefore supplies an attained positive closed-event pair with $a \ge 0$ in either case. By hypothesis, the full entry predicates hold at this least failure. The acquired-witness exclusion gives a contradiction. This proves all strict bounds above the base and hence all weak bounds.

Finally the independent Bernoulli limit (2) shows that the global supremum is at least C_{E}. Combined with the weak upper bounds, it gives the claimed equality. □

Under the same entry and external hypotheses, a base only through 41 has a different consequence: it locates a first failure, if one exists, in $42, \dots, 699$. An unresolved earlier failure would prevent us from supplying the sharp predecessor at a later index. The statement above therefore remains conditional on a genuine finite base, or on a separate complete argument closing that range. No unconditional cutoff at 700 is asserted.

8 Discussion and reproducibility

The mechanism of the selected route is now visible as one chain. Fixed-target extremality gives contact; integration gives a common physical budget; coarse entry makes that budget small. Direct moment geometry and two contact rays then acquire one strip, where the two noise comparisons exclude nonnegative excess.

The entry still contains two coarse passes, all four signed passes, 11264 nominal boxes and its internal 100-panel incidence contract. The direct strip argument replaces downstream localization and comparison branches only. The remaining global obligation is the finite base identified in Section 7.

Reproduction and its scope.

Sources, a dependency map and check summaries accompany this manuscript in PR 6 of `https://github.com/kolodziejek/berry-esseen-certificates`. The selected reference is PR 5 at commit `1aa1dd015e3d73fbeecc1a7acbfa395cbb560b5f`. Original analytical sources, numerical witnesses and the longer manuscript are preserved. Exact scalar checks and actual-law controls supplement the written proofs. They do not establish the missing finite base, and a successful document build is not numerical proof acceptance. The entry enclosures, their coverage and arithmetic semantics, and the stated external theorems remain explicit inputs.

AI assistance.

AI systems assisted extensively with research, programs, drafting, exposition and component review. Separate AI review roles examined interfaces between arguments. Their agreement is not mathematical verification or a substitute for independent human review. The authors remain responsible for the text and its claims. No full Lean formalization is claimed.

A The uniform Fourier enclosure

Fix an actual index $n \geq N \geq 42$, a centered variance-one marginal X and raw threshold H. Their values may vary with n: the estimates below are applied anew to that law and threshold, using common ceilings. In particular, no limiting law or comparison of characteristic functions at different indices is used. Write

$$f(t) = \mathbb{E}e^{itX}, \quad \gamma = \mathbb{E}X^3, \quad e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6), \quad V_n = f^n - e_n,$$
$$I_0 = \int_0^{1/2} \frac{|\Im(e^{-itH}V_n)|}{t^3}dt, \qquad I_\infty = \int_{1/2}^{\infty} \frac{|V_n|}{t^3}dt.$$

The phase e^{-itH} retains the raw threshold. At low frequency we use the imaginary part, whose cancellation is stronger than an absolute-value bound on V_n; on the tail an absolute-value bound is sufficient.

Compensated inversion.

Lemma A.1. *For a finite signed measure μ of mass and first moment zero with $\int x^2 d|\mu| < \infty$, every $H \in \mathbb{R}$ satisfies*

$$\int (x - H)|x - H|d\mu(x) = -\frac{4}{\pi}\int_0^\infty \frac{\Im(e^{-itH}\widehat{\mu}(t))}{t^3}dt,$$

with absolute convergence.

Proof. First take $y > 0$ and substitute $v = ty$. Integration by parts gives

$$\int_0^\infty \frac{ty - \sin ty}{t^3}dt = \frac{y^2}{2}\int_0^\infty \frac{1 - \cos v}{v^2}dv = \frac{\pi}{4}y^2.$$

The boundary term vanishes: $v - \sin v = O(v^3)$ at zero and $O(v)$ at infinity. To evaluate the constant, insert $v^{-2} = \int_0^\infty se^{-sv}ds$. Positive Tonelli gives

$$\int_0^\infty \frac{1 - \cos v}{v^2}dv = \int_0^\infty s\{s^{-1} - s/(s^2 + 1)\}ds = \pi/2.$$

Oddness extends the first identity to $\pi y|y|/4$ for every real y. Moreover $ty - \sin ty$ has the sign of y. Thus, for $y = x - H$, the absolute double integral against $d|\mu|(x)dt$ equals $\pi \int (x-H)^2 d|\mu|/4 < \infty$. Absolute Fubini is now justified. Since $\int (x-H)d\mu = 0$, the linear compensation disappears and gives the stated Fourier integral. This argument uses measures directly: atoms, including an atom at H, require no continuity convention. □

Apply this to the sum law minus the scale-$\sqrt{n}$ pushforward of $\phi(x)\{1 + \gamma(x^3 - 3x)/(6\sqrt{n})\}dx$. The comparator has mass one, mean zero and transform e_n. For $z = H/\sqrt{n}$, the identity $q_H(\sqrt{n}x) = nq_z(x)$ fixes the normalization. Three Gaussian integrations by parts, interpreted distributionally using $q_z''' = 4\delta_z$, give

$$\int q_z(x)(x^3 - 3x)\phi(x)dx = 4\phi(z).$$

Consequently the signed measure in the inversion lemma integrates q_H to $n\varepsilon$, with exactly the ε of the core identity. Multiplication by $-\sqrt{n}/2$ therefore gives its remainder $E_n = -n^{3/2}\varepsilon/2$ and the bound:

$$E_n = \frac{2\sqrt{n}}{\pi}\int_0^\infty \frac{\Im(e^{-itH}V_n)}{t^3}dt, \qquad |E_n| \le \frac{2\sqrt{n}}{\pi}(I_0 + I_\infty). \tag{59}$$

The complete low-frequency bound.

Assume common actual ceilings $\mu_4 \le M \le 111/10$, $|\gamma| \le g \le 37/20$, $\mu_6 \le M_6$, $|(3 - \mu_4)/24| \le A$, $|(10\gamma - \mathbb{E}X^5)/120| \le Z$. Put

$$E = M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \qquad B = A + Z/2 + E/4,$$
$$a_0 = (N-1)/(2N), \quad b_0 = (45 - M)(1 - 2/N)/96, \quad \Lambda_N = \sqrt{\pi/2}(N/(N-1))^{3/2}.$$

Proposition A.2. *For $c_0 \ge \Lambda_N$, $c_1 \ge \Lambda_N/2$,*

$$I_0 \le \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n},$$

where

$$L_0 = c_0 Z + c_1 Ag + 251g^3/17280, \quad L_H = c_0 A + 251g^2/4800,$$
$$L_2 = E/(2a_0^2) + B^2/(2b_0^3).$$

Proof. For $h = e^{-t^2/2 - i\gamma t^3/6}$ and $0 \le t \le 1/2$, Taylor's integral remainder gives

$$f - h = -\frac{3 - \mu_4}{24}t^4 - i\frac{10\gamma - \mathbb{E}X^5}{120}t^5 + r_6, \quad |r_6| \le Et^6, \quad |f - h| \le Bt^4. \tag{60}$$

For $w = -t^2/2 - i\gamma t^3/6$, use $|w| \le t^2(1/2 + g^2/144) < 1/6$ and $|e^w - 1 - w - w^2/2| \le |w|^3/5$; the sixth-degree term of $w^2/2$ costs $g^2t^6/72$, and the marginal remainder costs $\mu_6 t^6/720$. Four integrations of $1 - \cos u \ge 0$ give $\cos u \le 1 - u^2/2 + u^4/24$. Symmetrization implies

$$|f|^2 \le 1 - t^2 + (M+3)t^4/12 \le 1 - (45 - M)t^2/48.$$

Both f, h, and their segment, thus have modulus at most $e^{-(45-M)t^2/96}$. Taylor expansion of the polynomial v^n gives

$$f^n - h^n = n(f - h)h^{n-1} + R_2, \quad |R_2| \le \tfrac{1}{2}n(n-1)B^2t^8 e^{-(n-2)(45-M)t^2/96}.$$

Integrating r_6 and R_2 after division by t^3 costs L_2/n: $n-1 \ge 2a_0 n$, $(n-2)(45-M)/96 \ge b_0 n$, and $\int_0^\infty t^3 e^{-At^2}dt = 1/(2A^2)$, $\int_0^\infty t^5 e^{-At^2}dt = A^{-3}$. It remains to integrate the fourth- and fifth-order terms in (60); taking absolute values too early would lose the fourth-order cancellation. Put $b = n-1$. After rotation the real fourth-order coefficient contributes a sine of $Ht+b\gamma t^3/6$, whereas the fifth-order coefficient contributes a cosine. Hence their imaginary part is at most

$$ne^{-bt^2/2}\{(Z + A|H|)t^5 + bAgt^7/6\}.$$

After division by t^3, use

$$\int_0^\infty t^2 e^{-bt^2/2}dt = \sqrt{\pi/2}\, b^{-3/2}, \qquad \int_0^\infty t^4 e^{-bt^2/2}dt = 3\sqrt{\pi/2}\, b^{-5/2}.$$

These produce $\Lambda_n(Z + A|H| + Ag/2)/\sqrt{n}$. Since $n/(n-1) \le N/(N-1)$, this is bounded by the stated expression with Λ_N. Finally, putting $u = n\gamma t^3/6$ gives

$$|\Im(e^{-itH}(h^n - e_n))| \le e^{-nt^2/2}(|u|^3/6 + |H|tu^2/2).$$

Gaussian moments yield $\sqrt{\pi/2}\{5g^3/432+g^2|H|/24\}/\sqrt{n}$; $\sqrt{\pi/2} < 251/200$ proves the stated coefficients. All enlarged integrands are nonnegative. □

The sharper fifth-order input follows from an exact moment identity. For acquired support $|X| \le R \le 6$, $R \ge 1$, and $\mathbb{E}|X|^5 \le E_5$, take $M_6 = RE_5$ and $A = \max(2, M-3)/24$. For $Q = X^2 - \gamma X - 1$,

$$\mathbb{E}Q^2 = \mu_4 - 1 - \gamma^2, \qquad \mathbb{E}X^5 = \gamma^3 + 2\gamma + \mathbb{E}[Q^2(X + 2\gamma)].$$

Reflect this auxiliary moment calculation so $t = |\gamma|$. For $r \ge \max(R, 2g)$ both signs of $\mathbb{E}X^5 - 10\gamma$ are bounded by

$$P_r = r(M-1) + (10-2M)t - rt^2 + t^3, \quad Q_r = r(M-1) + (2M-10)t - rt^2 - t^3.$$

The coefficients of μ_4 are $r \mp 2t \ge 0$. If $M \le 5$, $P_r \ge Q_r$ and $P_r \le r(M-1) + (10-2M)t - (r-g)t^2$. If $M > 5$, $P_r \le r(M-1)$ and $Q_r \le r(M-1) + (2M-10)t - rt^2$. The quadratic maxima use vertices clipped to $[0, g]$. For $r = 6$, $P_6'' = -12 + 6t < 0$: when $M \le 5$ and $P_6'(g) \ge 0$ use $P_6(g)$; otherwise use $6(M-1) + (10-2M)^2/[4(6-g)]$. For $M > 5$ also use $6(M-1) + (2M-10)^2/24$. Minimizing the applicable ceilings with $E_5 + 10g$ and dividing by 120 therefore gives the claimed valid Z.

Every high-frequency piece and its floor gate.

For two iid copies assume an actual event $\mathcal{E}$ with $\mathbb{P}(\mathcal{E}) \ge \alpha > 0$, $D = |X - X'| \ge d > 0$ on $\mathcal{E}$, and $\mathbb{E}[D^2 \mid \mathcal{E}] \le M_E$. Fix $b = 9/10$, $0 < r \le 1/4$, and set

$$h = 333/106 - (22/7)^3 r^2/6 > 0, \quad \kappa = \alpha(23/48)^2 d^2, \quad L(x) = (x+2)/(x^2+4x+2),$$
$$S_3 = \sum_{j=1}^{20} j^{-3} + 1/800, \quad S_5 = \sum_{j=1}^{20} (j-r)^{-5} + [4(20-r)^4]^{-1},$$
$$c(u) = u/2 - (M+3)u^2/24, \quad c_* = \min\{c(1/4), c(81/100)\},$$
$$c_q = 1/8 - (M+3)/384, \qquad s = [1 - (M+3)53/600]/2.$$

Proposition A.3. *If $c_* > 0$, $\sqrt{n} I_\infty \le \sum_{i=1}^{6} T_i(n)$, where*

$$T_1 = 2\sqrt{n} e^{-nc_*}, \qquad T_2 = \frac{M_E}{4\pi^{3/2} h\sqrt{\alpha}} \{S_3 + 3S_5/(n\alpha h^2)\},$$

$$T_3 = \frac{M_E \sqrt{n}}{2\pi^2} e^{-n\alpha h^2 r^2}, \qquad T_4 = \frac{e^{-n\kappa b^2}}{2b^4 \kappa \sqrt{n}},$$

$$T_5 = \frac{\sqrt{n}}{2b^2} e^{-n\alpha (23/48)^2}, \qquad T_6 = \sqrt{n} e^{-n/8} \{2L(n/8) + \sqrt{M-1}/3\}.$$

The optional replacements, minimized with the respective simple terms, are

$$\widetilde{T}_1 = 2\sqrt{n} e^{-nc_q} L(ns/4) \quad (s > 0), \qquad \widetilde{T}_4 = \frac{\sqrt{n} e^{-n\kappa b^2} L(n\kappa b^2)}{2b^2}.$$

The common-floor sum at N is valid if

$$2Nc' > 1 \quad (c' \in \{c_*, \alpha h^2 r^2, \alpha (23/48)^2, 1/8\}).$$

The optional first term requires $2Nc_q > 1$, and the optional fourth $N\kappa b^2 \ge 1/4$. Without an optional gate, retain the simple term.

Proof. We first bound the characteristic function on $[1/2, b]$, then partition the remaining half-line around its possible lattice peaks. The comparator is integrated over the entire half-line separately. The six terms record these disjoint purposes: the initial interval, nonzero-cell cores and wings, two parts of the zero cell, and the comparator.

On $[1/2, b]$, $|f|^n \le e^{-nc(t^2)} \le e^{-nc_*}$ and $\int t^{-3} dt < 2$, giving T_1. Its optional chord follows from $c(u) - c_q - s(u - 1/4) = (M+3)(u - 1/4)(81/100 - u)/24 \ge 0$. For $x = nab^2 > 0$ the integration formula is

$$\sqrt{n} \int_b^\infty e^{-nat^2} t^{-3} dt \le \frac{\sqrt{n} e^{-x}}{2b^2} L(x). \tag{61}$$

Substitution and positive Fubini give the left side as this prefactor times $1 - xQ(x)$, where $Q(x) = \int_0^\infty e^{-u}/(x+u) du$. For $P(u) = u^2 - 4u + 2$ and $D_x = P(-x) = x^2 + 4x + 2$, use $P(u) - D_x = (u+x)(u - x - 4)$ and $\int_0^\infty P(u) e^{-u} du = \int_0^\infty u P(u) e^{-u} du = 0$. The two zero integrals follow from $\int_0^\infty u^k e^{-u} du = k!$. For clarity, the two divisions are

$$\int_0^\infty \frac{P(u) e^{-u}}{x+u} du = D_x Q(x) - x - 3,$$

$$\int_0^\infty \frac{P(u)^2 e^{-u}}{x+u} du = D_x \int_0^\infty \frac{P(u) e^{-u}}{x+u} du.$$

Indeed, the remainder in the second equality is the integral of $P(u)(u - x - 4)$, which vanishes by those two zero moments. Thus

$$Q(x) - \frac{x+3}{D_x} = \frac{1}{D_x^2} \int_0^\infty \frac{P(u)^2 e^{-u}}{x+u} du \ge 0.$$

This positive remainder proves $1 - xQ(x) \le L(x)$ for every $x > 0$, not just for large x. It proves (61); applying that formula to the chord majorant, extended to $[1/2, \infty)$, proves the optional first term.

For $t \ge b$, symmetrization and convex Jensen give

$$|f(t)|^n \le e^{-n\mathbb{E} \sin^2(t(X - X')/2)} \le \mathbb{E}[e^{-n\alpha \sin^2(tD/2)} \mid \mathcal{E}].$$

Here $|f|^2 = 1 - 2\mathbb{E}\sin^2(t(X - X')/2)$, so $1 - v \le e^{-v}$ first gives the exponential. Restricting the expectation to $\mathcal{E}$ and using its probability at least α then gives the conditional Jensen bound. Positive Tonelli permits a separate partition for each D: with $\ell = 2\pi/D$, use $[0, \ell/2]$ and nonzero cells $t = \ell(j + v)$, $j \ge 1$, $|v| \le 1/2$. This distance-dependent partition is made inside the conditional integral; it does not assume a common lattice for the original law. On $|v| \le r$, $|\sin \pi v| \ge h|v|$; on the wings it is at least hr, by $\sin x \ge x - x^3/6$ and monotonicity on $[0, \pi/2]$. Taylor's remainder gives

$$\{(j - v)^{-3} + (j + v)^{-3}\}/2 \le j^{-3} + 6(j - r)^{-5}v^2.$$

Pair v with $-v$ and extend each nonnegative Gaussian core integral to $\mathbb{R}$. With $a = n\alpha h^2$, its two required moments are $\int_{\mathbb{R}} e^{-av^2}dv = \sqrt{\pi/a}$ and $\int_{\mathbb{R}} v^2 e^{-av^2}dv = \sqrt{\pi}/(2a^{3/2})$. The Jacobian is $t^{-3}dt = D^2(j + v)^{-3}dv/(4\pi^2)$. After multiplying by $\sqrt{n}$ and averaging D^2, these factors give

$$\frac{M_E}{4\pi^{3/2}h\sqrt{\alpha}} \sum_{j \ge 1} \{j^{-3} + 3(j - r)^{-5}/(n\alpha h^2)\}.$$

For decreasing summands the tails beyond 20 are bounded by their integrals from 20 to infinity. These are $1/800$ and $[4(20 - r)^4]^{-1}$, respectively, proving T_2. The wing weight is at most $\int_{\ell/2}^{\infty} t^{-3}dt = D^2/(2\pi^2)$, giving T_3. Extending these sets below b only increases their bounds. In the zero cell, $u = tD/2 \le \pi/2$. On $u \le 1/2$, $\sin u \ge 23u/24$ and $D \ge d$. Integrate over the enlarged $[b, \infty)$ with $t^{-3} \le t/b^4$ to obtain T_4; (61) gives its optional form. On $u \ge 1/2$, $\sin u \ge 23/48$ gives T_5. Cell endpoints have zero Lebesgue measure even when D has atoms. For the comparator, $|e_n| \le e^{-nt^2/2}(1 + n|\gamma|t^3/6)$. Use (61) at $b = 1/2$ and $\int_{1/2}^{\infty} e^{-nt^2/2}dt \le 2e^{-n/8}/n$; $\gamma^2 \le M - 1$ from $\mathbb{E}(X^2 - \gamma X - 1)^2 \ge 0$ gives T_6. No comparator or lattice tail is omitted.

For floor evaluation, $\sqrt{n}e^{-nc'}$ decreases when $2nc' > 1$, whereas n^{-1} and $n^{-1/2}e^{-nc'}$ decrease for $c' > 0$. Also $L'(x) = -(x^2 + 4x + 6)/(x^2 + 4x + 2)^2 < 0$. The derivative of $\sqrt{x}e^{-x}L(x)$ has the sign of $4 - 10x - 22x^2 - 13x^3 - 2x^4$, negative and decreasing for $x \ge 1/4$. These prove every stated gate on a common scalar majorant, not by comparing characteristic functions of different laws. □

For any complete HIGH sum K_H at N, the final bound is

$$|E_n| \le \frac{2}{\pi}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\} \le \frac{16}{25}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\}. \tag{62}$$

Complete HIGH windows may be minimized; the LOW triple stays complete. Round denominator roots down and numerator roots up, using the constants below. For $x \ge 0$, choose rational $0 \le y \le \min(x, 64)$ and round $1/\sum_{j=0}^{64} y^j/j!$ upward to majorize e^{-x}. The floor proof concerns the analytical functions, not monotonicity of this rounded exponential helper.

B Elementary constants

Lemma B.1. *The standard normal density satisfies*

$$\|\phi''\|_\infty = \phi(0) < 2/5, \quad \|\phi'''\|_\infty \le 2/\pi, \quad 1/(4\pi) < 2/25,$$

$$819/2000 < C_{\mathrm{E}} < 20487/50000 < 41/100, \qquad \phi(0) > .39886 > .3988 = 997/2500.$$

Moreover $333/106 < \pi < 22/7$, $\sqrt{\pi} > 4431/2500$, *and* $0.4097321837023 < C_{\mathrm{E}} < 0.4097321837025$.

Proof. For $J(t) = \int e^{-x^2/2} e^{itx} dx$, integration by parts gives $J' = -tJ$; the squared Gaussian integral in polar coordinates gives $J(0) = \sqrt{2\pi}$. Thus $\phi(x) = (2\pi)^{-1} \int e^{-t^2/2} e^{-itx} dt$. Differentiating under integrable majorants gives $\|\phi'''\|_\infty \le (2\pi)^{-1} \int |t|^3 e^{-t^2/2} dt = 2/\pi$; likewise $\|\phi^{(4)}\|_\infty \le 3\phi(0)$ and $\|\phi^{(5)}\|_\infty \le 8/\pi$. The identity $\phi'' = (x^2 - 1)\phi$ gives its exact norm: the exterior maximum is $2e^{-3/2}\phi(0) < \phi(0)$.

Put $s_k(x) = \sum_{j=0}^{k-1} (-1)^j x^{2j+1}/(2j+1)$. Tangent addition and the bounds $0 < 4(1/5 - 1/(3 \cdot 5^3)) - 1/239 < 4\arctan(1/5) - \arctan(1/239) < 1 < \pi/2$ fix the first-quadrant branch (use $\pi/2 = 2\int_0^1 (1+t^2)^{-1} dt > 1$). Hence Machin's identity is $\pi = 16\arctan(1/5) - 4\arctan(1/239)$. The branch check is needed because tangent addition alone determines an angle only modulo π. All quantities in the lower and upper angle bounds above are elementary; the upper bound follows from $4\arctan(1/5) < 4/5 < 1$. For $0 < x < 1$, integrating the finite geometric identity gives the exact remainder

$$\arctan x - s_k(x) = (-1)^k \int_0^x \frac{t^{2k}}{1+t^2} dt.$$

Even k gives a lower bound and odd k an upper bound. Applying these opposite directions to the two terms of Machin's identity gives

$$\pi_- := 16 s_{12}(1/5) - 4 s_3(1/239) < \pi < 16 s_{11}(1/5) - 4 s_4(1/239) =: \pi_+.$$

Exact rational comparisons give $333/106 < \pi_- < \pi_+ < 22/7$ and $\pi_- > (4431/2500)^2$. Use the rational endpoints

$$r_- = 3162277660168379/10^{15}, \qquad r_+ = 3162277660168380/10^{15},$$

$$c_- = .4097321837023, \qquad c_+ = .4097321837025.$$

Squaring gives $r_-^2 < 10 < r_+^2$. Recall that $C_{\mathrm{E}}^2 = (19 + 6\sqrt{10})/(72\pi)$: its numerator increases with $\sqrt{10}$ and its denominator with π. Thus the lower enclosure uses r_- with π_+ and the upper one uses r_+ with π_-. Direct integer multiplication gives

$$c_-^2 < \frac{19 + 6r_-}{72\pi_+} < C_{\mathrm{E}}^2 < \frac{19 + 6r_+}{72\pi_-} < c_+^2.$$

This licenses the fine entry endpoints; the displayed coarse bounds for C_{E} follow since $819/2000 < c_- < c_+ < 20487/50000 < 41/100$. All denominators are positive and explicit. Finally $\pi > 25/8$ gives the coarse upper bounds, while $\pi < 22/7$ and $(.39886)^2 < 7/44$ give the lower bound for $\phi(0)$. □

References

[1] Andrew C. Berry. The accuracy of the Gaussian approximation to the sum of independent variates. *Transactions of the American Mathematical Society*, 49(1):122–136, 1941. doi: 10.1090/S0002-9947-1941-0003498-3.

[2] Carl-Gustav Esseen. On the Liapunoff limit of error in the theory of probability. *Arkiv för Matematik, Astronomi och Fysik*, 28A(9):1–19, 1942.

[3] Carl-Gustav Esseen. A moment inequality with an application to the central limit theorem. *Scandinavian Actuarial Journal*, 1956(2):160–170, 1956. doi: 10.1080/03461238.1956.10414946.

[4] Jona Schulz. *The optimal Berry–Esseen constant in the binomial case*. PhD thesis, Universität Trier, 2016. URL https://ubt.opus.hbz-nrw.de/frontdoor/index/index/docId/732.

[5] Irina G. Shevtsova. On the absolute constants in the Berry–Esseen inequality and its structural and nonuniform improvements. *Informatics and its Applications*, 7(1):124–125, 2013. URL https://www.mathnet.ru/eng/ia252. In Russian.

[6] Vladimir M. Zolotarev. An absolute estimate of the remainder term in the central limit theorem. *Theory of Probability and its Applications*, 11(1):95–105, 1966. doi: 10.1137/1111005.